

Unit 2:

Basic Mathematical Operations on Vectors

What We'll Cover

- 1. Matrix Math with Vectors:** Dive into the basics of vector math! We'll practice adding, subtracting, scaling, and multiplying vectors using matrices.
- 2. Connecting Geometry with Matrix Math:** See how geometric ideas (like moving and stretching vectors) connect to operations in matrix algebra.

3. Do Matrix Math with Vectors:

- Add and subtract vectors using their components or matrix forms.
- Multiply vectors by scalars and understand how it changes their size or direction.

4. Dive into Matrix Products:

- Calculate the dot product of vectors and understand what it tells you (like angles or projections).
- Learn about the cross product (if we're working in 3D) and what it means geometrically.

5. Tackle Real-World Problems:

- Solve practical problems using vectors and matrix operations, whether it's geometry or real-life examples.

Basic Mathematical Operations on Vectors

1. Scalar Multiplication of Vectors

A scalar is a real number, such as 2, 0.033, 10.5, and so on. Multiplying a vector by a scalar is called **scalar multiplication**. To perform scalar multiplication, each component of the vector is multiplied by the scalar. The resulting vector has the same direction as the original, but a magnitude scaled by the scalar.

Given a vector $\mathbf{A} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix}$ and a scalar λ , scalar multiplication is defined as:

$$\lambda \mathbf{A} = \begin{bmatrix} \lambda a_x \\ \lambda a_y \\ \lambda a_z \end{bmatrix}.$$

Worked Example

A physical vector **A**, described by the matrix

$$\mathbf{A} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

is scaled by a factor of 2.

- (a) Calculate the scaled vector.
- (b) Draw the two vectors and comment about their size and direction.

Solution

Given:

$$\mathbf{A} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \lambda = 2.$$

The scaled vector $\mathbf{A}'$ is:

$$\mathbf{A}' = \lambda \mathbf{A} = 2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \cdot 2 \\ 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

The resulting scaled vector is:

$$\mathbf{A}' = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

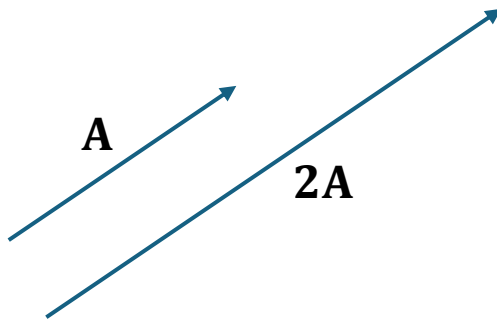


Figure 14

The scaled vector ($2\mathbf{A}$) is twice as long as vector $\mathbf{A}$. Both vectors point in the same direction, as the scalar multiplication by 2 preserves the direction of the original vector. The result is a vector with a magnitude twice that of $\mathbf{A}$, visually represented by its increased length while maintaining alignment with $\mathbf{A}$.

Worked Example

A physical vector **B**, described by the matrix

$$\mathbf{B} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

is scaled by a factor of $-\frac{1}{2}$.

- (a) Calculate the scaled vector.
- (b) Draw the two vectors and comment about their size and direction.

Solution

Given:

$$\mathbf{B} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \lambda = -\frac{1}{2}.$$

The scaled vector $\mathbf{B}'$ is:

$$\mathbf{B}' = \lambda \mathbf{A} = -\frac{1}{2} \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \cdot 4 \\ -\frac{1}{2} \cdot 6 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$$

The resulting scaled vector is:

$$\mathbf{B}' = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$$

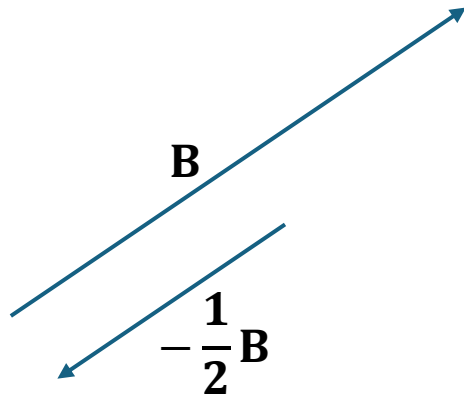


Figure 15

The scalar multiple $-\frac{1}{2}\mathbf{B}$ has a magnitude that is half of the magnitude of the original vector $\mathbf{B}$. The negative scalar reverses the direction of the vector, so $-\frac{1}{2}\mathbf{B}$ points in the opposite direction to $\mathbf{B}$.

Properties of Scalar Multiplication

- Scalar multiplication satisfies many of the same properties as the usual multiplication.
 1. $\lambda(\mathbf{A} + \mathbf{B}) = \lambda\mathbf{A} + \lambda\mathbf{B}$ (Distributive law, Form 1)
 2. $(\lambda + \beta)\mathbf{A} = \lambda\mathbf{A} + \beta\mathbf{A}$ (Distributive law, Form 2)
 3. $1\mathbf{A} = \mathbf{A}$
 4. $(-1)\mathbf{A} = -\mathbf{A}$
 5. $0\mathbf{A} = \mathbf{0}$

2. Addition of Vectors

Vector addition is the operation of adding two or more vectors together to obtain a vector sum. Geometrically, the vector sum of two position vectors, $\overrightarrow{OP}$ and $\overrightarrow{OQ}$, is obtained by placing the vectors head-to-tail and drawing the resulting vector ($\overrightarrow{OR}$) from the free tail to the free head (see Figure 16).

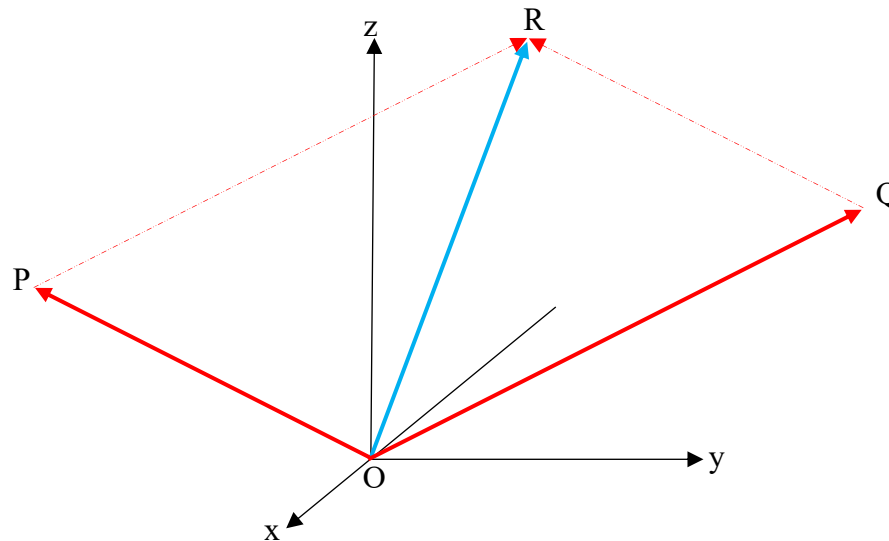


Figure 16

Symbolically, we write the vector sum of $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ that results in the vector $\overrightarrow{OR}$ as:

$$\overrightarrow{OR} = \overrightarrow{OP} + \overrightarrow{OQ}$$

If position vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ are expressed in column matrix form as:

$$\overrightarrow{OP} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}, \quad \overrightarrow{OQ} = \begin{bmatrix} q_x \\ q_y \\ q_z \end{bmatrix}$$

then,

$$\overrightarrow{OR} = \overrightarrow{OP} + \overrightarrow{OQ} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} + \begin{bmatrix} q_x \\ q_y \\ q_z \end{bmatrix} = \begin{bmatrix} p_x + q_x \\ p_y + q_y \\ p_z + q_z \end{bmatrix} \blacksquare$$

Worked Example 2

Given two position vectors **A** and **B** in column matrix form:

$$\mathbf{A} = \begin{bmatrix} 3 \\ 12 \\ 6 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -4 \\ -3 \\ 2 \end{bmatrix}, \quad \text{find the components of } \mathbf{A} + \mathbf{B}.$$

Notes

1. When single-letter labels such as **A** and **B** are used for position vectors, they typically represent the directed line segments from the origin **O** to the points **A** and **B**, respectively. In contrast, when not explicitly defined as position vectors, these labels could represent physical vectors, which do not necessarily originate from the origin of the coordinate system.
2. The rules of vector addition also applies to physical vectors.

Solution

To find the components of $\mathbf{A} + \mathbf{B}$, simply add the corresponding components of the two position vectors.

Given:

$$\mathbf{A} = \begin{bmatrix} 3 \\ 12 \\ 6 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -4 \\ -3 \\ 2 \end{bmatrix}$$

Add them component-wise:

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 3 + (-4) \\ 12 + (-3) \\ 6 + 2 \end{bmatrix}$$

Thus, the components of $\mathbf{A} + \mathbf{B}$ are: $\begin{bmatrix} -1 \\ 9 \\ 8 \end{bmatrix}$

Vector Addition: Concepts and Properties

Vector addition satisfies two important properties:

1. The Commutative Law:

This law states that the order of addition does not matter. In mathematical terms,

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}.$$

This property is also referred to as the **parallelogram law**. As illustrated in Figure 17, two adjacent edges of the parallelogram represent $\mathbf{A} + \mathbf{B}$ and $\mathbf{B} + \mathbf{A}$. However, both sums are equal in magnitude and direction, represented by the diagonal of the parallelogram.

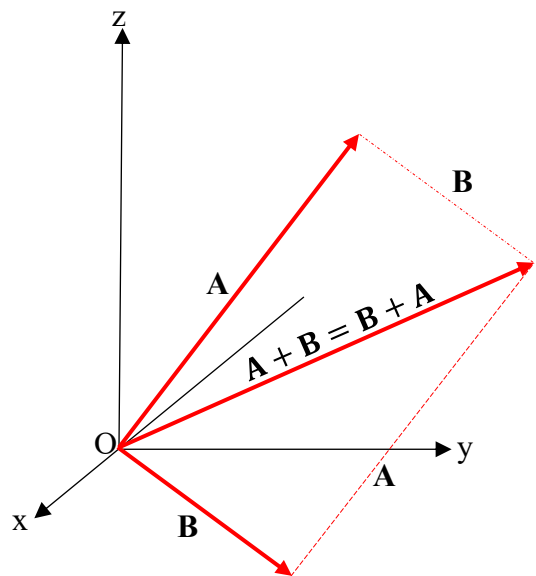


Figure 17

2. The Associative Law:

This law states that the sum of three vectors remains the same regardless of the grouping of the vectors during addition. Mathematically, this can be expressed as:

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}).$$

Additionally, the associative property allows flexibility in rearranging the grouping:

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = (\mathbf{A} + \mathbf{C}) + \mathbf{B},$$

and

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = (\mathbf{C} + \mathbf{A}) + \mathbf{B}$$

Application of Scalar Multiplication and Vector Addition

Application 1: Finding the Midpoint between Two Points in a 3-D System

Finding the midpoint of a pair of points in 3-D space is straight forward. Imagine you have two points with position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ (see Figure 18). To find the midpoint, you simply average their position vectors. That is, if $\mathbf{r}_m$ is the position vector of the midpoint, then

$$\mathbf{r}_m = \frac{\mathbf{r}_1 + \mathbf{r}_2}{2}$$

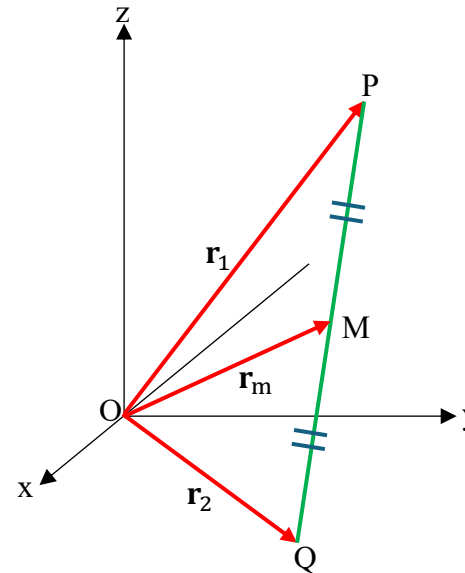


Figure 18

Worked Example

Given the position vectors of two points, P and Q, respectively, as:

$$\mathbf{r}_1 = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix} \text{ m}, \quad \mathbf{r}_2 = \begin{bmatrix} -3 \\ 2 \\ 4 \end{bmatrix} \text{ m}.$$

What is the position vector of the midpoint between the two points. Show a plot of them in a 3-D coordinate system.

Solution

The midpoint position vector $\mathbf{r}_m$ is the average of $\mathbf{r}_1$ and $\mathbf{r}_1$:

$$\mathbf{r}_m = \frac{\mathbf{r}_1 + \mathbf{r}_2}{2}$$

Using matrix addition and scalar multiplication:

$$\begin{aligned}\mathbf{r}_m &= \frac{\mathbf{r}_1 + \mathbf{r}_2}{2} = \frac{1}{2} [\mathbf{r}_1 + \mathbf{r}_2] \\ &= \frac{1}{2} \left[\begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix} + \begin{bmatrix} -3 \\ 2 \\ 4 \end{bmatrix} \right] = \frac{1}{2} \begin{bmatrix} 2 + (-3) \\ -1 + 2 \\ 5 + 4 \end{bmatrix}\end{aligned}$$

Therefore,

$$\mathbf{r}_m = \begin{bmatrix} -0.5 \\ 0.5 \\ 4.5 \end{bmatrix}$$

Figure 19 shows a 3-D plot of points P, Q, and midpoint.

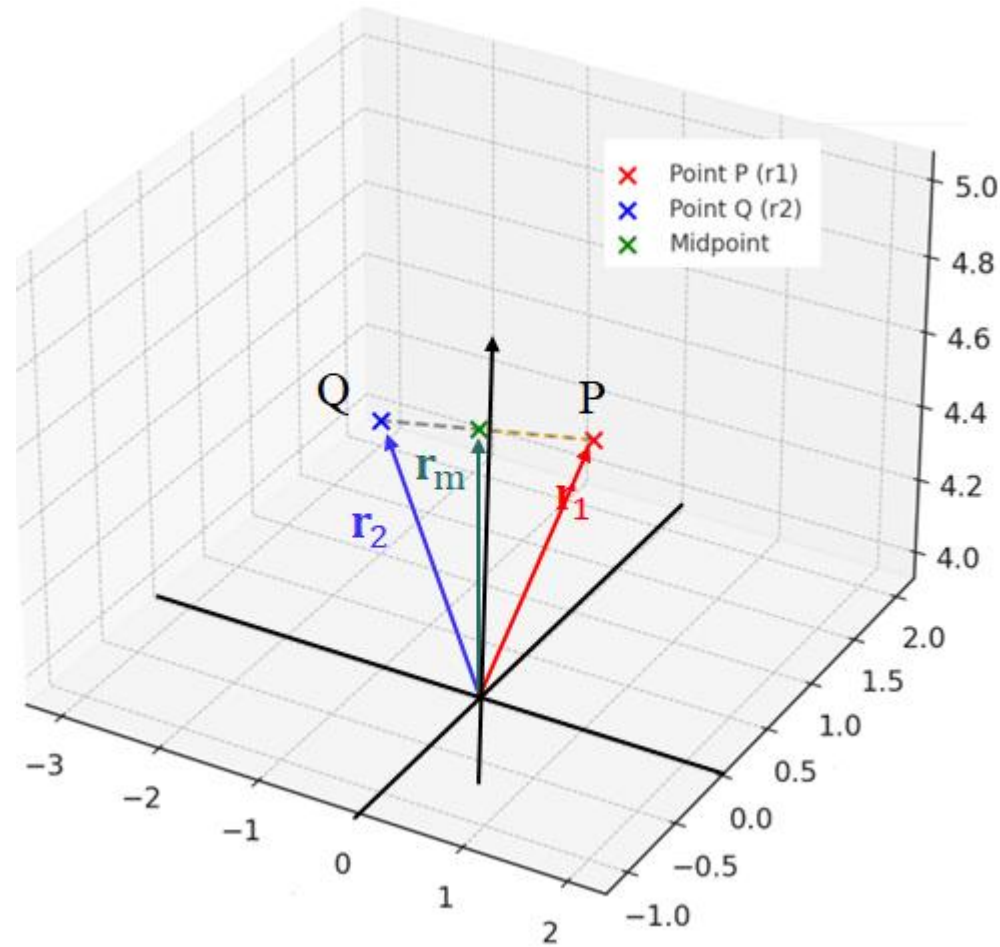


Figure 19

Applications of Scalar Multiplication and Vector Addition

Application 2 : Finding the Centre of Mass of a System of Particles.

The centre of mass (COM) is a fundamental concept in physics that **describes the average position of the mass of an object or a system of objects**. It simplifies the study of the motion and dynamics of complex systems by treating them as a single point particle located at the centre of mass.

Quantitatively, the **centre of mass (COM)** of a system of particles is the weighted average of their positions, where the weights are their masses. For a system of n particles with masses $m_1, m_2, \dots, m_n$ and positions $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n$ (see Figure 20), the centre of mass is defined as:

$$\mathbf{r}_{\text{cm}} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2 + \dots + m_n \mathbf{r}_n}{m_1 + m_2 + \dots + m_n}$$

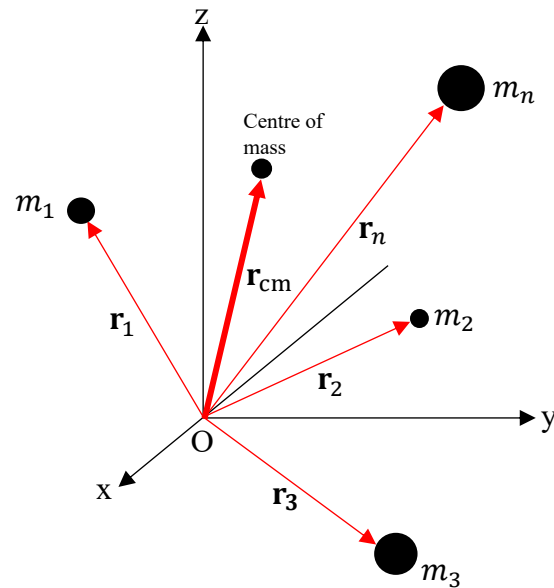


Figure 20: Centre of mass of particles

Worked Example

A system of two particles has masses and position vectors as follows:

- Sphere 1 has mass $m_1 = 5$ kg and position vector $\mathbf{r}_1 = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$ m,
- Sphere 2 has mass $m_2 = 10$ kg and position vector $\mathbf{r}_2 = \begin{bmatrix} 8 \\ -35 \\ 2 \end{bmatrix}$ m.

Find the centre of mass $\mathbf{r}_{cm}$.

Solution

For a system of two particles in 3D space (see Figure 21), the position of the center of mass is the point on the line that joins the two particles where the total mass of the system is considered to be concentrated. The COM can be calculated using the formula

$$\mathbf{r}_{\text{cm}} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2}$$

- m_1 and m_2 are the masses of the two particles
- $\mathbf{r}_1$ and $\mathbf{r}_2$ are the position vectors of the two masses.

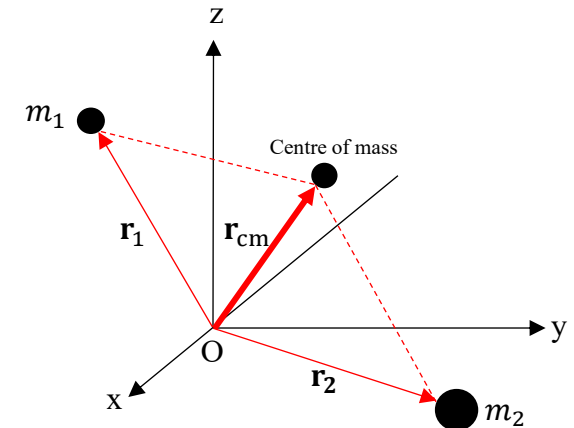


Figure 21

Given:

$$m_1 = 5 \text{ kg},$$

$$\mathbf{r}_1 = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \text{ m}$$

$$m_2 = 10 \text{ kg},$$

$$\mathbf{r}_2 = \begin{bmatrix} 8 \\ -3 \\ 2 \end{bmatrix} \text{ m}$$

Substitute into the formula:

$$\mathbf{r}_{\text{cm}} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2}$$

We get:

$$\mathbf{r}_{\text{cm}} = \frac{1}{(5+10)} \left[5 \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} + 10 \begin{bmatrix} 8 \\ -3 \\ 2 \end{bmatrix} \right] = \begin{bmatrix} 6 \\ -1 \\ 1 \end{bmatrix} \text{ m} \blacksquare$$

3. Subtraction of Vectors

Before defining vector subtraction, we first introduce the concept of the vector $(-\mathbf{A})$, which represents the opposite of vector $\mathbf{A}$. The vector $(-\mathbf{A})$ has the same magnitude as $\mathbf{A}$, but points in the opposite direction. Geometrically, we can visualize the vectors as being parallel to each other but pointing in opposite directions (see Figure 22).

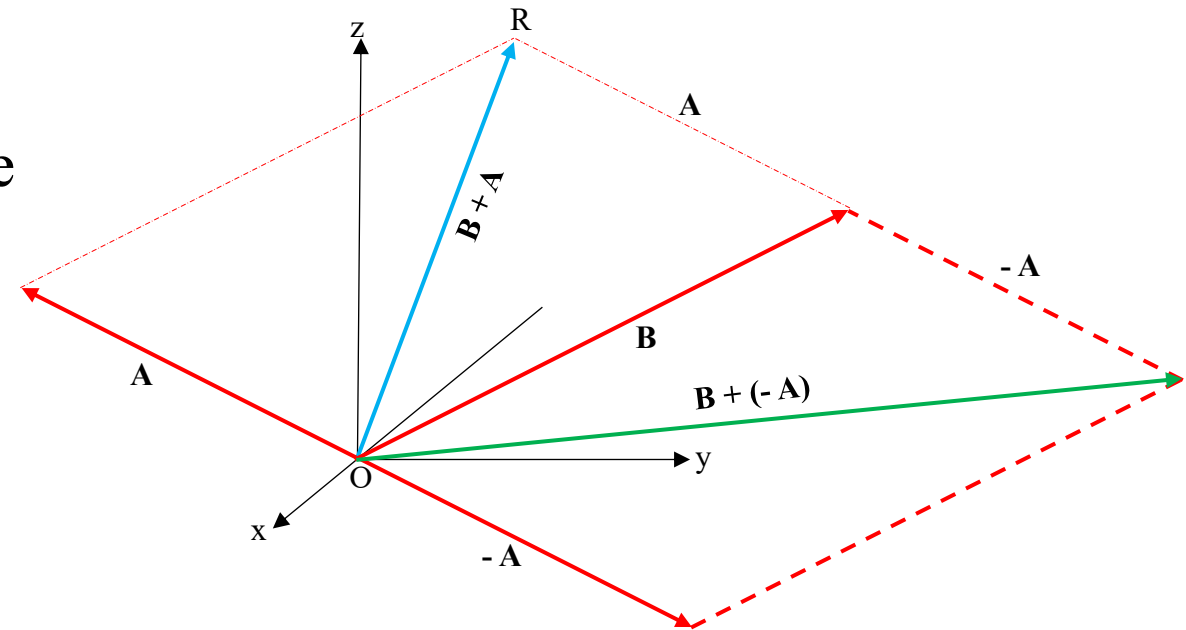


Figure 22

With this understanding, we define the subtraction of vector **A** from vector **B** as the addition of **B** and the opposite of **A**. Mathematically, this is expressed as:

$$\mathbf{B} - \mathbf{A} = \mathbf{B} + (-\mathbf{A})$$

Geometrically, this operation is equivalent to reversing the direction of vector **A** and then applying the rules of vector addition.

Suppose we are given physical vectors, **A** and **B**, where:

$$\mathbf{A} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix}$$

The vector subtraction of **B** from **A** that results in vector **C** is written as:

$$\mathbf{C} = \mathbf{B} - \mathbf{A}$$

In column matrix form, we have

$$\mathbf{C} = \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} - \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \begin{bmatrix} b_x - a_x \\ b_y - a_y \\ b_z - a_z \end{bmatrix} \blacksquare$$

Application of Vector Subtraction: Finding Distance Between Two Points

If two points, P and Q, have coordinates (p_x, p_y, p_z) and (q_x, q_y, q_z) , respectively, then the **displacement vector** (or **separation vector**) pointing from P to Q (see Figure 23), is labelled as $\overrightarrow{PQ}$ and is the difference between their position vectors.

That is,

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$$

In column vector form, we write

$$\overrightarrow{PQ} = \begin{bmatrix} q_x \\ q_y \\ q_z \end{bmatrix} - \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} = \begin{bmatrix} q_x - p_x \\ q_y - p_y \\ q_z - p_z \end{bmatrix}$$

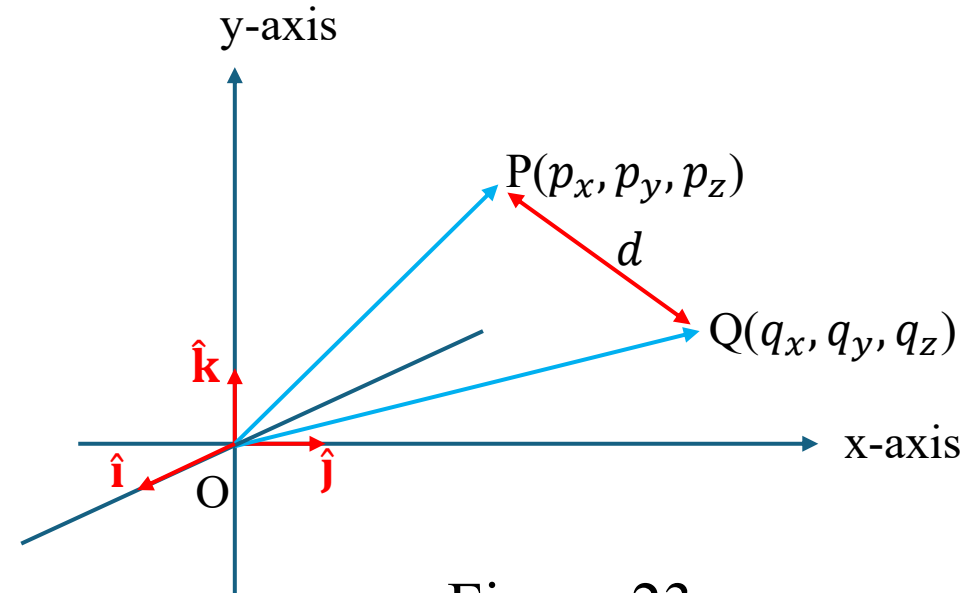


Figure 23

The magnitude of vector $\overrightarrow{PQ}$ gives the distance between P and Q. That is,

$$\text{Distance, } d = |\overrightarrow{PQ}| = \sqrt{(q_x - p_x)^2 + (q_y - p_y)^2 + (q_z - p_z)^2}$$

Worked Example

A particle moves from position $P(3, -2, 0)$ to position $Q(5, -8, -4)$. Find the distance vector $\overrightarrow{PQ}$ and calculate the distance travelled by the particle.

Solution

Position vectors of the two points are:

$$\overrightarrow{OP} = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix}, \quad \overrightarrow{OQ} = \begin{bmatrix} 5 \\ -8 \\ -4 \end{bmatrix}$$

(see Figure 24)

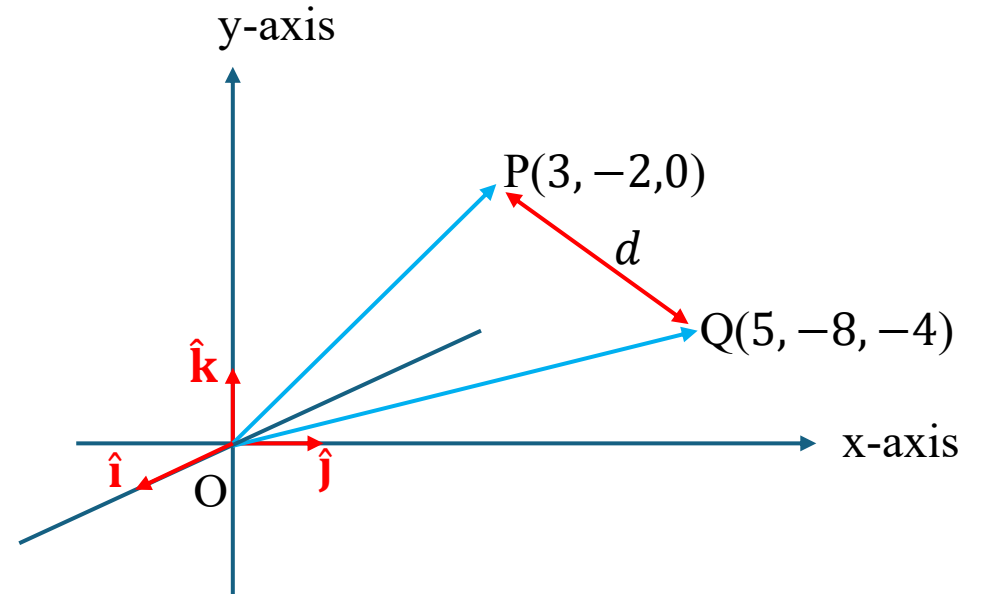


Figure 24

The displacement vector from point P to point Q is:

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \begin{bmatrix} 5 \\ -8 \\ -4 \end{bmatrix} - \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix}$$

Upon performing the subtraction, we get:

$$\overrightarrow{PQ} = \begin{bmatrix} 5 - 3 \\ -8 + 2 \\ -4 - 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ -4 \end{bmatrix}$$

The distance between the two points is given by the magnitude of the vector $\overrightarrow{PQ}$, which is calculated as:

$$\text{Distance, } d = |\overrightarrow{PQ}| = \sqrt{2^2 + (-6)^2 + (-4)^2} = \sqrt{56} \blacksquare$$

Worked Example

A system of three point charges has the following charges and their respective locations on a 2D rectangular coordinate system:

Charge $q_1 = 8 \mu\text{C}$, located at coordinates (0 m, 0.12 m)

Charge $q_2 = -6 \mu\text{C}$, located at coordinates (0 m, 0 m)

Charge $q_3 = 10 \mu\text{C}$, located at coordinates (0.15 m, 0 m)

Using Coulomb's law, determine the resultant force acting on charge q_1 due to charges q_2 and q_3 .

Solution

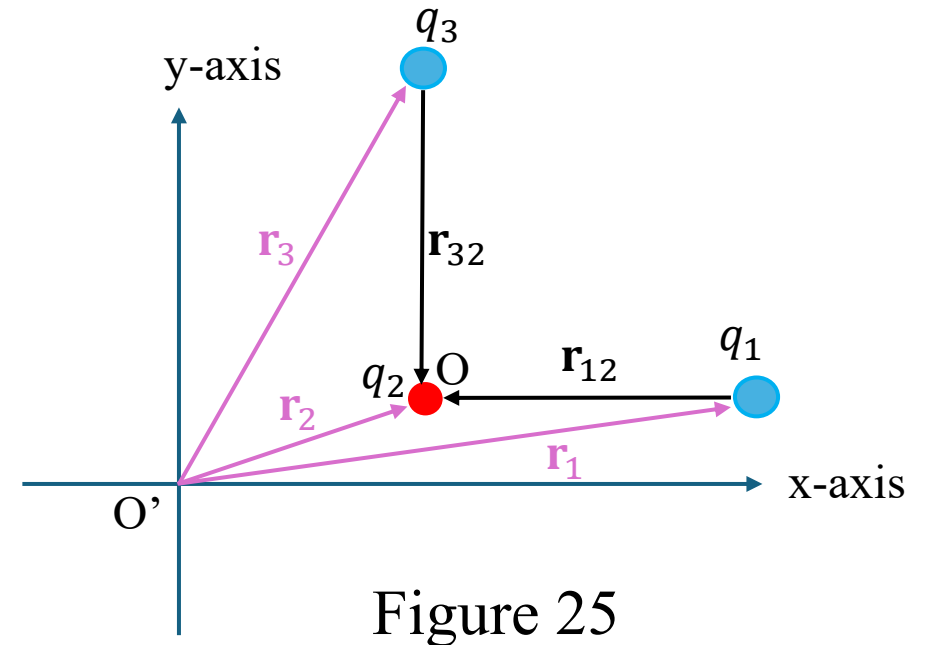
Step 1: Compute the position vectors of the charges.

If the observation point is denoted as O' (see Figure 25), then the position vectors of the charges can be expressed as follows:

- $q_1 = 8 \mu\text{C}$: $\mathbf{r}_1 = \langle 0.0, 0.12 \rangle \text{ m}$
- $q_2 = -6 \mu\text{C}$: $\mathbf{r}_2 = \langle 0.0, 0.0 \rangle \text{ m}$
- $q_3 = -2 \mu\text{C}$: $\mathbf{r}_3 = \langle 0.7, 0.5 \rangle \text{ m}$
- $q_4 = 10 \mu\text{C}$: $\mathbf{r}_4 = \langle 0.15, 0 \rangle \text{ m}$

The Coulomb constant is taken as:

$$k = 9 \times 10^9 \text{ N/m}^2 \cdot \text{C}$$



Step 2: Compute the displacement vectors.

Let all displacement vectors be defined as pointing toward charge q_2 . This convention ensures consistency when determining the directions of attractive and repulsive forces for vectorial addition. Note that in 2D or 3D systems, forces must be resolved into components before performing vector addition. Alternatively, displacement vectors can be defined as pointing away from q_2 , provided this choice is applied consistently.

For example, if the displacement vector from charge q_3 to charge q_2 is labeled as $\mathbf{r}_{32}$, it is given by:

$$\mathbf{r}_{32} = \mathbf{r}_2 - \mathbf{r}_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0.15 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.15 \\ 0 \end{bmatrix} \text{ m}$$

Similarly, if we label the displacement vector from charge q_1 to charge q_2 as $\mathbf{r}_{12}$, then, we have:

$$\mathbf{r}_{12} = \mathbf{r}_2 - \mathbf{r}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 0.12 \end{bmatrix} = \begin{bmatrix} 0 \\ -0.12 \end{bmatrix} \text{ m}$$

Step 3: Calculate the coulomb force on charge q_2 for each charge:

- Force on charge q_2 due to charge q_1 :

$$\mathbf{F}_{12} = k \frac{q_1 q_2}{|\mathbf{r}_{12}|^3} \mathbf{r}_{12}$$

Substituting:

$$\mathbf{F}_{12} = 9 \times 10^9 \frac{(8 \times 10^{-6})(-6 \times 10^{-6})}{(\sqrt{0^2 + 0.12^2})^3} \begin{bmatrix} 0 \\ -0.12 \end{bmatrix} \text{ N}$$

Computing the force vector $\mathbf{F}_{12}$:

$$\mathbf{F}_{12} = \begin{bmatrix} 0 \\ 30 \end{bmatrix} \text{ N}$$

Step 3: Calculate the coulomb force on charge q_2 for each charge (continued)

- Force on charge q_2 due to charge q_3 :

$$\mathbf{F}_{32} = k \frac{q_3 q_2}{|\mathbf{r}_{12}|^3} \mathbf{r}_{32}$$

Substituting:

$$\mathbf{F}_{32} = 9 \times 10^9 \frac{(10 \times 10^{-6})(-6 \times 10^{-6})}{(\sqrt{0^2 + 0.15^2})^3} \begin{bmatrix} -0.15 \\ 0 \end{bmatrix} \text{ N}$$

Computing the force vector $\mathbf{F}_{32}$:

$$\mathbf{F}_{32} = \begin{bmatrix} 24 \\ 0 \end{bmatrix} \text{ N}$$

Step 4: Find the Resultant Force

The net force is the vector sum:

$$\mathbf{F}_{\text{net}} = \mathbf{F}_{12} + \mathbf{F}_{32}$$

Substitute the vectors:

$$\mathbf{F}_{\text{net}} = \begin{bmatrix} 0 \\ 30 \end{bmatrix} + \begin{bmatrix} 24 \\ 0 \end{bmatrix} = \begin{bmatrix} 24 \\ 30 \end{bmatrix} \text{ N} \blacksquare$$

The signs of the component forces indicate that charge q_1 exerts an attractive force on charge q_2 in the positive y-direction, while charge q_3 exerts an attractive force on charge q_2 in the positive x-direction. Together, these forces result in a net force in the first quadrant (positive x- and y-directions).

Let's solve another problem to practice assigning correct directions for displacement vectors and calculating forces.

Problem:

Four point charges are placed at the corners of a square as shown in Figure 26:

- $q_1 = q_3 = -2 \mu\text{C}$
- $q_2 = q_4 = 5 \mu\text{C}$

Using Coulomb's law, determine the resultant force acting on charge q_1 due to charges q_2 , q_3 and q_4 .

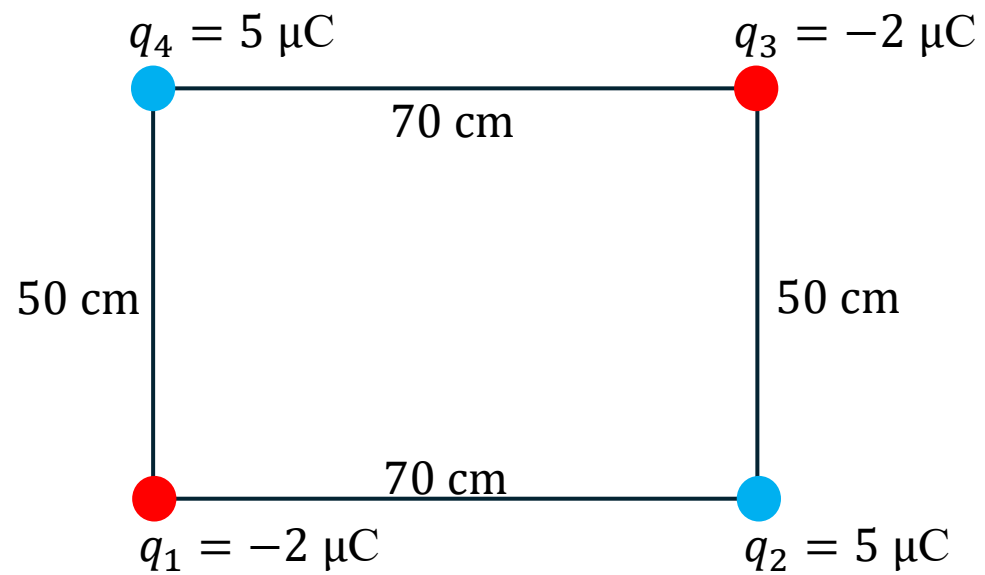


Figure 26

Solution

We begin by setting up the system of charges on a 2-D rectangular coordinate plane.

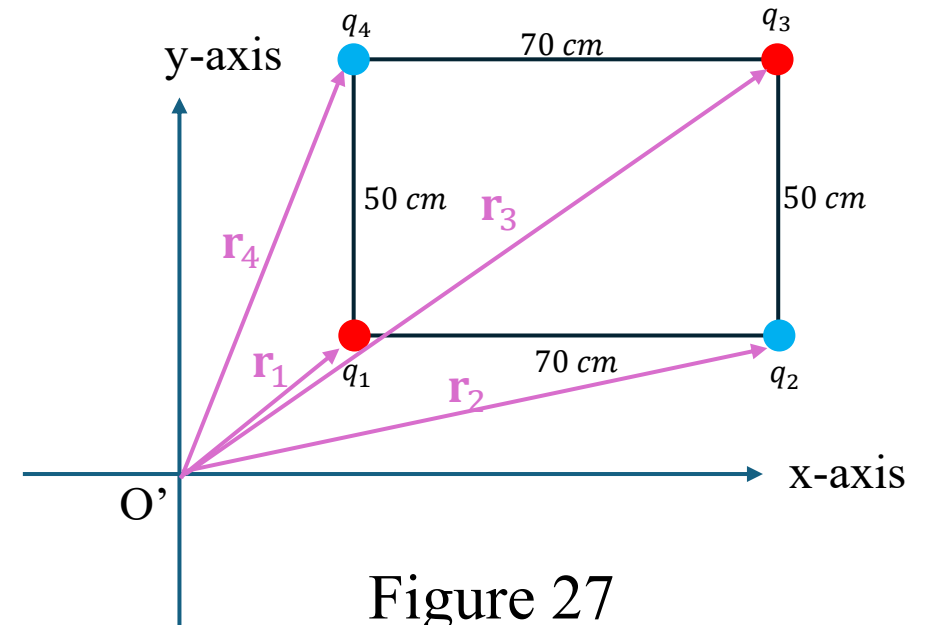
Place charge q_1 directly at the origin. If the observation point is denoted as O' (see

Figure 27), then the position vectors of the charges can be expressed as follows:

- $q_1 = -2 \mu\text{C}$: $\mathbf{r}_1 = \langle 0.0, 0.0 \rangle \text{ m}$
- $q_2 = 5 \mu\text{C}$: $\mathbf{r}_2 = \langle 0.7, 0.0 \rangle \text{ m}$
- $q_3 = -2 \mu\text{C}$: $\mathbf{r}_3 = \langle 0.7, 0.5 \rangle \text{ m}$
- $q_4 = 5 \mu\text{C}$: $\mathbf{r}_4 = \langle 0.0, 0.5 \rangle \text{ m}$

The Coulomb constant is taken as:

$$k = 9 \times 10^9 \text{ N/m}^2 \cdot \text{C}$$



Steps

1. Compute the displacement vectors.

- Displacement vector from charge q_2 to charge q_1 :

$$\mathbf{r}_{21} = \mathbf{r}_1 - \mathbf{r}_2 = \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix} - \begin{bmatrix} 0.7 \\ 0.0 \end{bmatrix} = \begin{bmatrix} -0.7 \\ 0.0 \end{bmatrix} \text{ m}$$

- Displacement vector from charge q_3 to charge q_1 :

$$\mathbf{r}_{31} = \mathbf{r}_1 - \mathbf{r}_3 = \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix} - \begin{bmatrix} 0.7 \\ 0.5 \end{bmatrix} = \begin{bmatrix} -0.7 \\ -0.5 \end{bmatrix} \text{ m}$$

- Displacement vector from charge q_4 to charge q_1 :

$$\mathbf{r}_{41} = \mathbf{r}_1 - \mathbf{r}_4 = \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix} - \begin{bmatrix} 0.0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 0.0 \\ -0.5 \end{bmatrix} \text{ m}$$

2. Calculate the coulomb force on charge q_1 for each charge:

- Force on charge q_1 due to charge q_2 :

$$\mathbf{F}_{21} = k \frac{q_1 q_2}{|\mathbf{r}_{21}|^3} \mathbf{r}_{21}$$

Substituting:

$$\mathbf{F}_{21} = 9 \times 10^9 \frac{(-2 \times 10^{-6})(5 \times 10^{-6})}{(\sqrt{(0.0)^2 + (-0.7)^2})^3} \begin{bmatrix} -0.7 \\ 0.0 \end{bmatrix} \text{N}$$

Computing the force vector $\mathbf{F}_{12}$:

$$\mathbf{F}_{21} = \begin{bmatrix} 0.184 \\ 0 \end{bmatrix} \text{N} \quad (\text{check})$$

2. Calculate the coulomb force on charge q_1 for each charge: (continued)

- Force on charge q_2 due to charge q_3 :

$$\mathbf{F}_{31} = k \frac{q_3 q_1}{|\mathbf{r}_{31}|^3} \mathbf{r}_{31}$$

Substituting:

$$\mathbf{F}_{31} = 9 \times 10^9 \frac{(-2 \times 10^{-6})(-2 \times 10^{-6})}{(\sqrt{(-0.7)^2 + (-0.5)^2})^3} \begin{bmatrix} -0.7 \\ -0.5 \end{bmatrix} \text{ N}$$

Computing the force vector $\mathbf{F}_{31}$:

$$\mathbf{F}_{31} = \begin{bmatrix} -0.04 \\ -0.028 \end{bmatrix} \text{ N}$$

2. Calculate the coulomb force on charge q_1 for each charge: (continued)

- Force on charge q_1 due to charge q_4 :

$$\mathbf{F}_{41} = k \frac{q_4 q_1}{|\mathbf{r}_{41}|^3} \mathbf{r}_{41}$$

Substituting:

$$\mathbf{F}_{41} = 9 \times 10^9 \frac{(5 \times 10^{-6})(-2 \times 10^{-6})}{(\sqrt{0^2 + (-0.5)^2})^3} \begin{bmatrix} 0.0 \\ -0.5 \end{bmatrix} \text{ N}$$

Computing the force vector $\mathbf{F}_{41}$:

$$\mathbf{F}_{41} = \begin{bmatrix} 0.0 \\ 0.36 \end{bmatrix} \text{ N}$$

3. Find the Resultant Force

The net force is the vector sum:

$$\mathbf{F}_{\text{net}} = \mathbf{F}_{21} + \mathbf{F}_{31} + \mathbf{F}_{41}$$

Substitute the vectors:

$$\mathbf{F}_{\text{net}} = \begin{bmatrix} 0.184 \\ 0 \end{bmatrix} + \begin{bmatrix} -0.04 \\ -0.028 \end{bmatrix} + \begin{bmatrix} 0.0 \\ 0.36 \end{bmatrix} = \begin{bmatrix} 0.144 \\ 0.332 \end{bmatrix} \text{ N} \blacksquare$$

The signs of the component forces indicate that charge q_2 exerts an attractive force on charge q_1 in the positive y-direction, charge q_3 exerts a repulsive force on charge q_1 in the negative y-direction, and charge q_4 exerts an attractive force on charge q_1 in the positive x-direction. Together, these forces result in a net force in the first quadrant (positive x- and y-directions).

Worked Example

Four charges are arranged in a square with sides of length 2.5 cm (see Figure 28). The two charges in the top right and bottom left corners are $+3.0 \times 10^{-6}$ C. The charges in the other two corners are -3.0×10^{-6} C. What is the net force exerted on the charge in the top right corner by the other three charges?

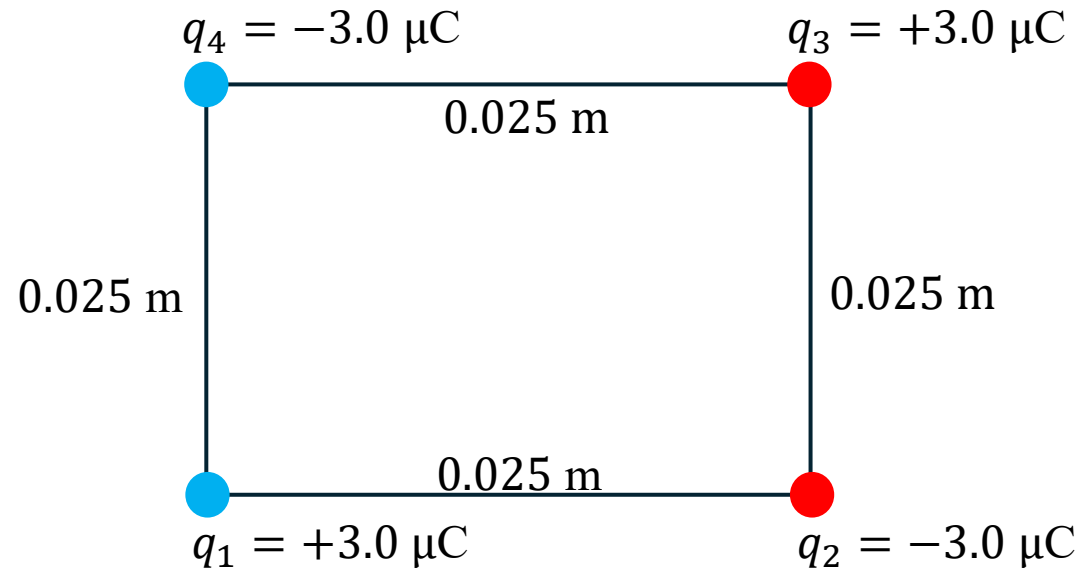


Figure 28

If the observation point is denoted as O' (see Figure 29), then the position vectors of the charges can be expressed as follows:

- $q_1 = 3.0 \mu\text{C}$: $\mathbf{r}_1 = \langle 0.0, 0.0 \rangle \text{ m}$
- $q_2 = -3.0 \mu\text{C}$: $\mathbf{r}_2 = \langle 0.025, 0.0 \rangle \text{ m}$
- $q_3 = 3.0 \mu\text{C}$: $\mathbf{r}_3 = \langle 0.025, 0.025 \rangle \text{ m}$
- $q_4 = -3.0 \mu\text{C}$: $\mathbf{r}_4 = \langle 0.0, 0.025 \rangle \text{ m}$

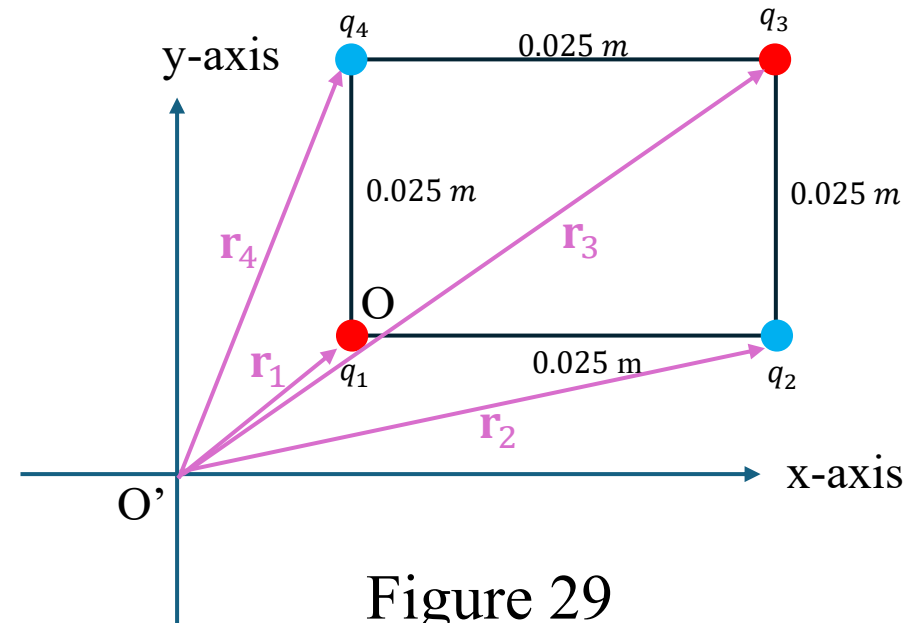


Figure 29

Steps

1. Compute the displacement vectors (see Figure 30).

- Displacement vector from charge q_1 to charge q_3 :

$$\mathbf{r}_{13} = \mathbf{r}_3 - \mathbf{r}_1 = \begin{bmatrix} 0.025 \\ 0.025 \end{bmatrix} - \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix} = \begin{bmatrix} 0.025 \\ 0.025 \end{bmatrix} \text{ m}$$

- Displacement vector from charge q_2 to charge q_3 :

$$\mathbf{r}_{23} = \mathbf{r}_3 - \mathbf{r}_2 = \begin{bmatrix} 0.025 \\ 0.025 \end{bmatrix} - \begin{bmatrix} 0.025 \\ 0.0 \end{bmatrix} = \begin{bmatrix} 0.0 \\ 0.025 \end{bmatrix} \text{ m}$$

- Displacement vector from charge q_4 to charge q_3 :

$$\mathbf{r}_{43} = \mathbf{r}_3 - \mathbf{r}_4 = \begin{bmatrix} 0.025 \\ 0.025 \end{bmatrix} - \begin{bmatrix} 0.0 \\ 0.025 \end{bmatrix} = \begin{bmatrix} 0.025 \\ 0.0 \end{bmatrix} \text{ m}$$

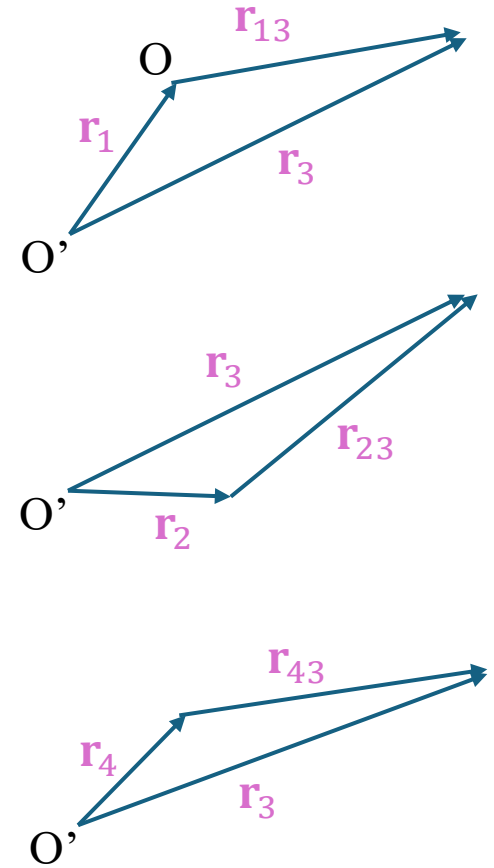


Figure 30

2. Calculate the coulomb force on charge q_3 for each charge:

- Force on charge q_3 due to charge q_1 :

$$\mathbf{F}_{13} = k \frac{q_1 q_3}{|\mathbf{r}_{13}|^3} \mathbf{r}_{13}$$

Substituting:

$$\mathbf{F}_{13} = 9 \times 10^9 \frac{(3.0 \times 10^{-6})(3.0 \times 10^{-6})}{(\sqrt{(0.025)^2 + (0.025)^2})^3} \begin{bmatrix} 0.025 \\ 0.025 \end{bmatrix} \text{N}$$

Computing the force vector $\mathbf{F}_{13}$:

$$\mathbf{F}_{13} = \begin{bmatrix} 45.82 \\ 45.82 \end{bmatrix} \text{N} \quad |\mathbf{F}_{13}| = 64.8 \text{ N}$$

2. Calculate the coulomb force on charge q_3 for each charge:

- Force on charge q_3 due to charge q_2 :

$$\mathbf{F}_{23} = k \frac{q_2 q_3}{|\mathbf{r}_{23}|^3} \mathbf{r}_{23}$$

Substituting:

$$\mathbf{F}_{23} = 9 \times 10^9 \frac{(-3.0 \times 10^{-6})(3.0 \times 10^{-6})}{\left(\sqrt{(0.0)^2 + (0.025)^2}\right)^3} \begin{bmatrix} 0.0 \\ 0.025 \end{bmatrix} \text{N}$$

Computing the force vector $\mathbf{F}_{23}$:

$$\mathbf{F}_{23} = \begin{bmatrix} 0.0 \\ -129.6 \end{bmatrix} \text{N} \quad |\mathbf{F}_{23}| = 129.6 \text{ N}$$

2. Calculate the coulomb force on charge q_3 for each charge:

- Force on charge q_3 due to charge q_4 :

$$\mathbf{F}_{43} = k \frac{q_4 q_3}{|\mathbf{r}_{43}|^3} \mathbf{r}_{43}$$

Substituting:

$$\mathbf{F}_{43} = 9 \times 10^9 \frac{(-3.0 \times 10^{-6})(3.0 \times 10^{-6})}{\left(\sqrt{(0.025)^2 + (0.0)^2}\right)^3} \begin{bmatrix} 0.025 \\ 0.0 \end{bmatrix} \text{N}$$

Computing the force vector $\mathbf{F}_{23}$:

$$\mathbf{F}_{43} = \begin{bmatrix} -129.6 \\ 0.0 \end{bmatrix} \text{N} \quad |\mathbf{F}_{43}| = 129.6 \text{ N}$$

3. Find the Resultant Force

The net force is the vector sum:

$$\mathbf{F}_{\text{net}} = \mathbf{F}_{13} + \mathbf{F}_{23} + \mathbf{F}_{43}$$

Substitute the vectors:

$$\mathbf{F}_{\text{net}} = \begin{bmatrix} 45.82 \\ 45.82 \end{bmatrix} + \begin{bmatrix} 0.0 \\ -129.6 \end{bmatrix} + \begin{bmatrix} -129.6 \\ 0.0 \end{bmatrix} = \begin{bmatrix} -83.79 \\ -83.79 \end{bmatrix} \text{ N} \blacksquare$$

giving,

$$|\mathbf{F}_{\text{net}}| = 118 \text{ N}$$

The result is a net force of 118 N directed downward along the diagonal of the square.

Dot Product of Two Vectors

The **dot product** is a mathematical operation that quantifies the degree to which two vectors align with each other. It provides a measure of how much one vector projects onto another. The result of the dot product is always a **scalar quantity**, meaning it has magnitude but no direction.

The dot product of two vectors, **A** and **B**, that results in the scalar quantity **W** is written symbolically as:

$$W = \mathbf{A} \cdot \mathbf{B}$$

This operation is widely used in engineering and physics to analyze forces, work, and directional relationships between vectors.

Geometric Interpretation of the Dot Product

The dot product of two physical vectors, **A** and **B** (see Figure 31), is also related to the angle θ between them through the formula:

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\theta$$

Here, $|\mathbf{A}|$ and $|\mathbf{B}|$ represent the magnitudes (or lengths) of the vectors, and θ is the angle between them.

This relationship allows us to determine the angle between two vectors by rearranging the formula:

$$\cos\theta = \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|}$$

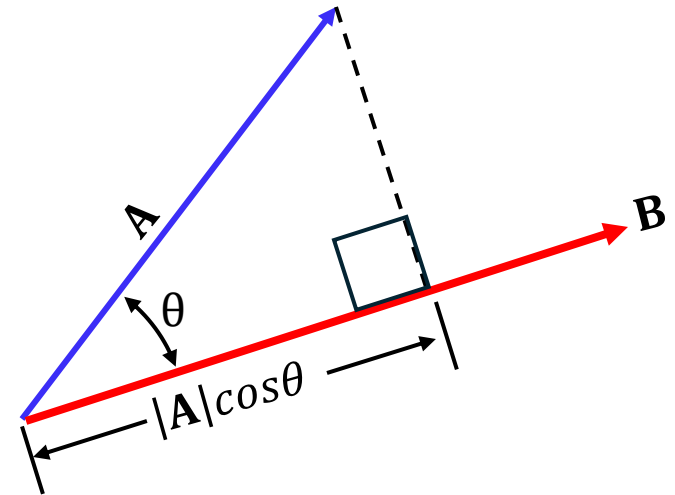


Figure 31

Matrix Notation of the Dot Product

In matrix notation, to compute the dot product of two vectors **A** and **B**, each containing n elements, we proceed as follows:

1. **Represent the vectors as column matrices:**

$$\mathbf{A} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

2. **Transpose the first vector to convert it into a row vector:**

$$\mathbf{A}^T = [a_1 \quad a_2 \quad \cdots \quad a_n]$$

3. Perform matrix multiplication between the transposed vector and the second vector:

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{A}^T \mathbf{B} = [a_1 \quad a_2 \quad \cdots \quad a_n] \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \sum_{i=1}^n a_i b_i$$

Note

It's important to note that directly multiplying two column vectors is undefined in matrix multiplication because their dimensions are incompatible. By transposing the first vector into a row vector, we align the dimensions appropriately, allowing for a valid matrix multiplication that results in the dot product.

Example:

Let's take two 3-dimensional column vectors:

$$\mathbf{A} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

The dot product of $\mathbf{A}$ and $\mathbf{B}$ is:

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{A}^T \mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$= 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6$$

$$= 32$$

Example

Given two vectors:

$$\mathbf{A} = \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 4 \\ 5 \end{bmatrix} \quad (\text{see Figure 32})$$

Compute the angle θ between them.

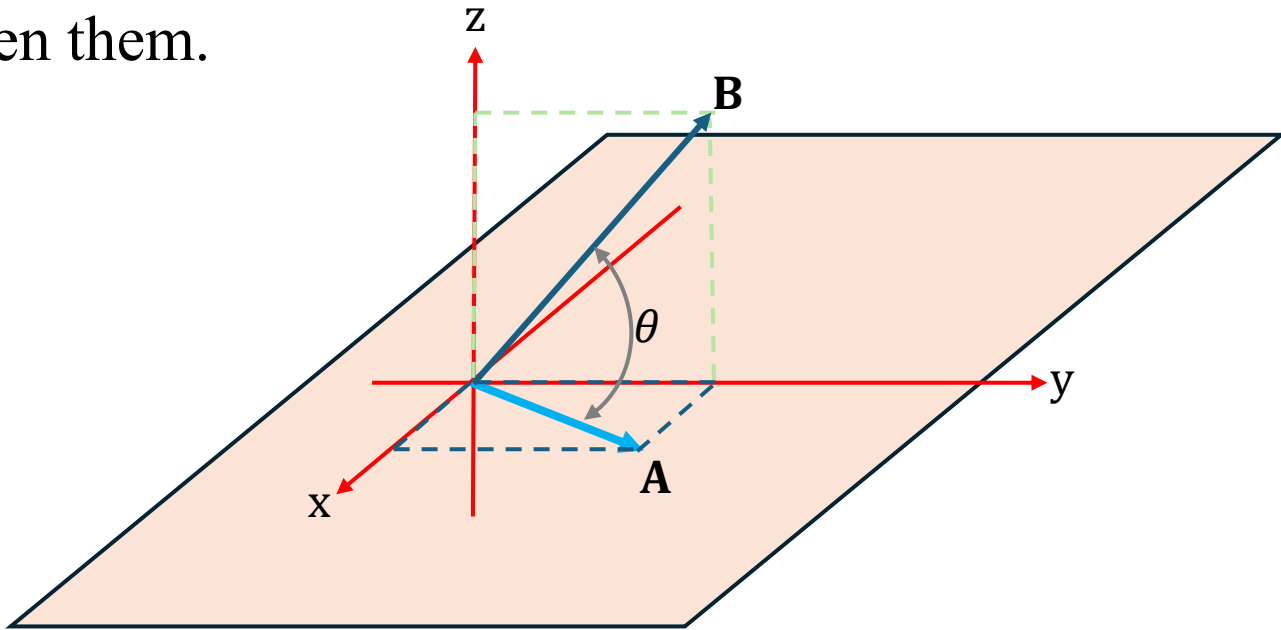


Figure 32

Solution

Step 1: Calculate the Dot Product $\mathbf{A} \cdot \mathbf{B}$:

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} &= \mathbf{A}^T \mathbf{B} = \begin{bmatrix} 3 & 4 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 4 \\ 5 \end{bmatrix} \\ &= 3 \cdot 0 + 4 \cdot 4 + 0 \cdot 5 = 16\end{aligned}$$

Step 2: Calculate the magnitudes of $\mathbf{A}$ and $\mathbf{B}$

$$|\mathbf{A}| = \sqrt{3^2 + 4^2 + 0} = 5$$

$$|\mathbf{B}| = \sqrt{0 + 4^2 + 5^2} = \sqrt{41}$$

Step 3: Substitute into the Cosine Formula

$$\cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|} = \frac{16}{5 \times \sqrt{41}} = 0.5$$

Step 4: Calculate the Angle θ

To find θ , we take the inverse cosine:

$$\theta = \cos^{-1}(0.5) = 60^\circ$$

Result:

The angle between vectors $\mathbf{A} = \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 \\ 4 \\ 5 \end{pmatrix}$ is 60° .

Applications of the Dot Product in Engineering Problem Solving

1. Calculating the Mechanical Work Done by a Force

The **work done (W)** by a force **F** (see Figure 33) pushing on an object moving through a displacement **d** is given by the dot product formula:

$$W = \mathbf{F} \cdot \mathbf{d} = |\mathbf{F}||\mathbf{d}|\cos\theta$$

where:

- **F** is the applied force vector,
- **d** is the displacement vector,
- θ is the angle between **F** and **d**.

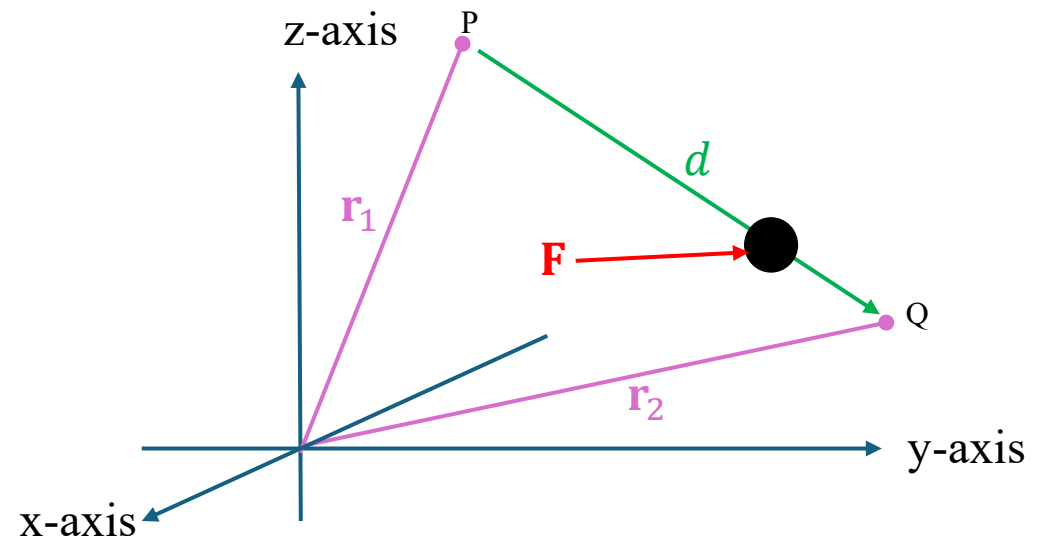


Figure 33

Worked Example 1: Pulling a Heavy Object Along a Floor with Friction

Given Data:

- A force of **50 N** is applied at an angle of **30°** to pull a mass (see Figure 34).
- The displacement of the mass is 10 meters along the x-axis.
- Represent the force and displacement as vectors.

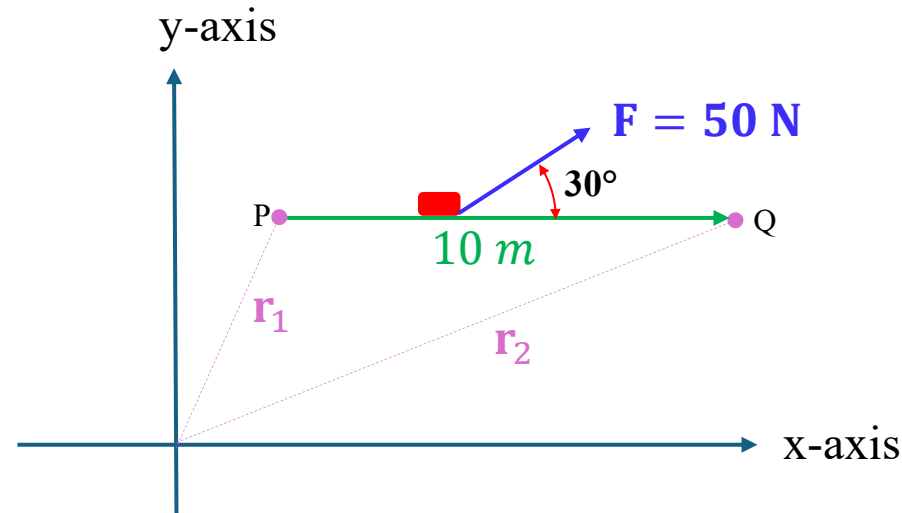


Figure 34

Solution

Step 1: Expressing Vectors as Matrices

The force vector $\mathbf{F}$ and displacement vector $\mathbf{d}$ in component form are:

$$\mathbf{F} = \begin{bmatrix} F_x \\ F_y \end{bmatrix} = \begin{bmatrix} 50 \cdot \cos 30^\circ \\ 50 \cdot \sin 30^\circ \end{bmatrix} = \begin{bmatrix} 43.2 \\ 23 \end{bmatrix} \text{ N}$$

$$\mathbf{d} = \begin{bmatrix} d_x \\ d_y \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \end{bmatrix} \text{ m}$$

Step 2: Using Matrix Dot Product to Calculate Work Done

The dot product of two column vectors $\mathbf{F}$ and $\mathbf{d}$ is calculated as:

$$\begin{aligned} W &= \mathbf{F}^T \cdot \mathbf{d} = [43.2 \quad 25] \begin{bmatrix} 10 \\ 0 \end{bmatrix} \\ &= (43.2 \times 10) + (25 \times 0) \\ W &= 433 \text{ J} \end{aligned}$$

Thus, the work done by the force is 433 joules.

Worked Example 1: Work Done Pulling a Heavy Object Along a Floor with Friction

Given Data:

- A force of **200 N** is applied at an angle of **15°** to pull an object (see Figure 35).
- The object moves **25 meters** along the x-axis.

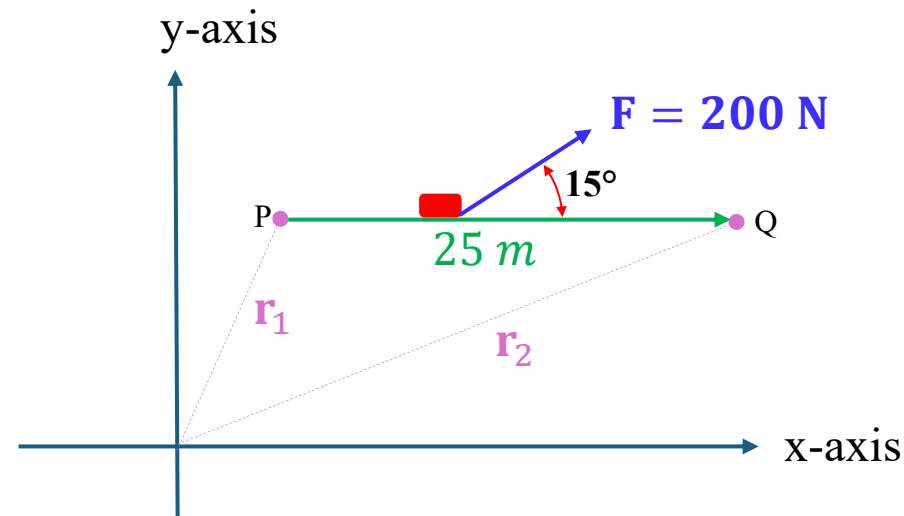


Figure 35

Solution

Step 1: Force and Displacement Vectors

$$\mathbf{F} = \begin{bmatrix} F_x \\ F_y \end{bmatrix} = \begin{bmatrix} 200 \cdot \cos 15^\circ \\ 200 \cdot \sin 15^\circ \end{bmatrix} = \begin{bmatrix} 193.2 \\ 51.8 \end{bmatrix} \text{ N}$$

$$\mathbf{d} = \begin{bmatrix} d_x \\ d_y \end{bmatrix} = \begin{bmatrix} 25 \\ 0 \end{bmatrix} \text{ m}$$

Step 2: Work Done Calculation

$$W = \mathbf{F}^T \mathbf{d}$$

$$W = [193.2 \quad 51.8] \begin{bmatrix} 25 \\ 0 \end{bmatrix}$$

$$= (193.2 \times 25) + (51.8 \times 0)$$

$$W = 4830 \text{ J}$$

Thus, the work done is **4830 joules**.

Properties of Dot Product

The **dot product**, also known as the **scalar product**, has several important properties that make it useful in engineering, physics, and mathematics. Here are the key properties of the dot product:

1. Commutative Property

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$$

- The order of the vectors does not affect the result.
- Example: If $\mathbf{A} = [2, 4]$ and $\mathbf{B} = [4, 1]$ then:

$$\mathbf{A} \cdot \mathbf{B} = (2 \cdot 4) + (3 \cdot 1) = 8 + 3 = 11$$

$$\mathbf{B} \cdot \mathbf{A} = (3 \cdot 1) + (2 \cdot 4) = 3 + 8 = 11$$

2. Distributive Property

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$$

- The dot product distributes over vector addition.
- Example: If $\mathbf{B} = [2, 4]$ and $\mathbf{C} = [3, 4]$ then:

$$\mathbf{B} + \mathbf{C} = [4, 6]$$

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot [4, 6]$$

Separately:

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C} &= [(2 \cdot 1) + (3 \cdot 2) + (2 \cdot 3) + (3 \cdot 4)] \\ &= 8 + 18 \\ &= 26\end{aligned}$$

3. Scalar Multiplication Property

$$(c\mathbf{A}) \cdot \mathbf{B} = c(\mathbf{A} \cdot \mathbf{B})$$

- A scalar multiple of a vector can be factored out of the dot product.
- Example: If $\mathbf{A} = [2, 3]$, $\mathbf{B} = [1, 2]$ and $c = 2$,

$$(2\mathbf{A}) \cdot \mathbf{B} = [4, 6] \cdot [1, 2] = (4 \cdot 1) + (6 \cdot 2) = 16$$

$$2(\mathbf{A} \cdot \mathbf{B}) = 2[(2 \cdot 1) + (3 \cdot 2)] = 16$$

4. Scalar Multiplication Property

$$\mathbf{A} \cdot \mathbf{0} = 0$$

- The dot product of any vector with the zero vector is always zero.
- Example: If $\mathbf{A} = [5, 7]$,

$$\mathbf{A} \cdot [0, 0] = (5 \cdot 0) + (7 \cdot 0) = 0$$

5. Orthogonal Property

$\mathbf{A} \cdot \mathbf{B} = 0$ if and only if $\mathbf{A} \perp \mathbf{B}$

- If two vectors are perpendicular (orthogonal), their dot product is zero.
- Example: Vectors $[3, 4]$ and $[-4, 3]$ are perpendicular because:

$$(3 \cdot -4) + (4 \cdot 3) = -12 + 12 = 0$$

6. Magnitude (Norm) Property

$$\mathbf{A} \cdot \mathbf{A} = |\mathbf{A}|^2$$

The dot product of a vector with itself gives the square of its magnitude.

- Example: If $\mathbf{A} = [3, 4]$,

$$\mathbf{A} \cdot \mathbf{A} = (3 \cdot 3) + (4 \cdot 4) = 9 + 16 = 25$$

$$|\mathbf{A}|^2 = \sqrt{25} = 5$$

7. Angle Property Interpretation

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos \theta$$

- The dot product relates to the cosine of the angle between two vectors.
- Example: If $|\mathbf{A}| = 5$, $|\mathbf{B}| = 10$, and the angle $\theta = 60^\circ$:

$$\mathbf{A} \cdot \mathbf{B} = 5 \times 10 \times \cos 60^\circ = 50 \times 0.5 = 25$$

8. Positivity Property

$$\mathbf{A} \cdot \mathbf{A} \geq 0$$

- The dot product of a vector with itself is always non-negative (zero only if the vector is the zero vector).

Note

These properties make the dot product a powerful tool in engineering applications such as force analysis, work calculations, and vector projections.

Dot Products of Unit Vectors

The dot products of unit vectors in the Cartesian coordinate system (rectangular coordinate system) are fundamental to vector operations. In 3D space, the standard unit vectors are:

$$\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$$

These unit vectors represent the directions along the x, y, and z axes, respectively (see Figure 36).

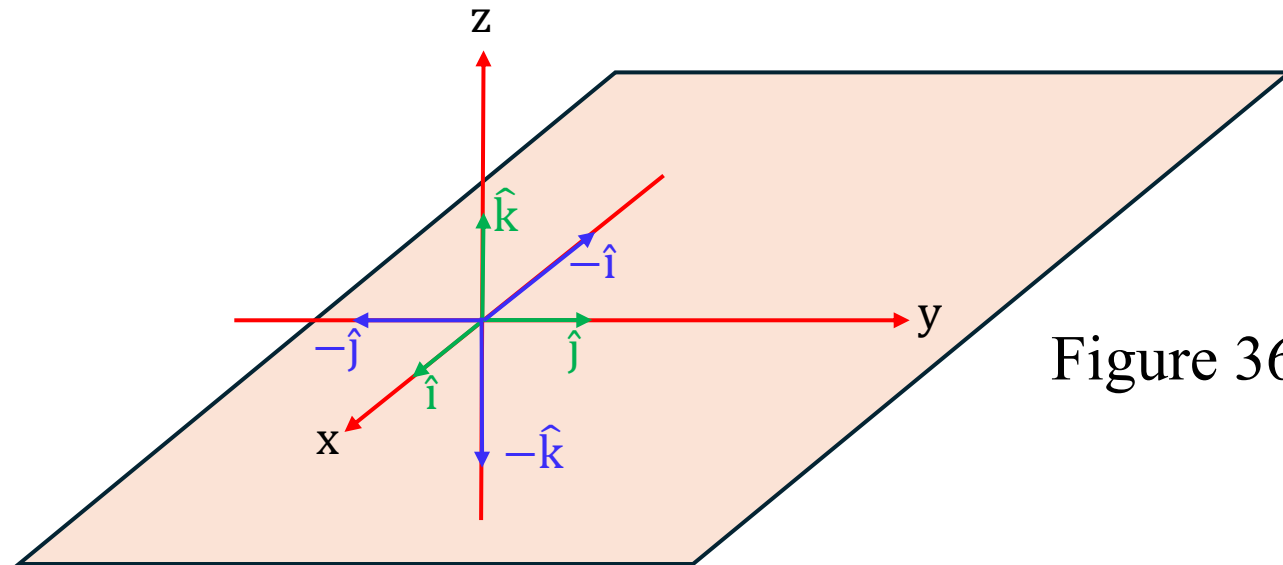


Figure 36

1. Dot Product of a Unit Vector with Itself (Magnitude Squared):

$$\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 1, \quad \hat{\mathbf{j}} \cdot \hat{\mathbf{j}} = 1, \quad \hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 1$$

- Each unit vector has a magnitude of 1.
- The dot product of a unit vector with itself results in 1, because the angle between them is 0° (cosine of 0° is 1).

2. Dot Product of Different Unit Vectors (Orthogonality):

$$\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = 0, \quad \hat{\mathbf{j}} \cdot \hat{\mathbf{k}} = 0, \quad \hat{\mathbf{k}} \cdot \hat{\mathbf{i}} = 0$$

- Different unit vectors are perpendicular to each other.
- The dot product of perpendicular vectors is zero, because the angle between them is 90° (cosine of 90° is 0).

Summary in Matrix Form (Dot Product Table):

$$\begin{bmatrix} \hat{\mathbf{i}} \cdot \hat{\mathbf{i}} & \hat{\mathbf{i}} \cdot \hat{\mathbf{j}} & \hat{\mathbf{i}} \cdot \hat{\mathbf{k}} \\ \hat{\mathbf{j}} \cdot \hat{\mathbf{i}} & \hat{\mathbf{j}} \cdot \hat{\mathbf{j}} & \hat{\mathbf{j}} \cdot \hat{\mathbf{k}} \\ \hat{\mathbf{k}} \cdot \hat{\mathbf{i}} & \hat{\mathbf{k}} \cdot \hat{\mathbf{j}} & \hat{\mathbf{k}} \cdot \hat{\mathbf{k}} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This is the identity matrix, which confirms that unit vectors are orthogonal and normalized.

Example Using Dot Product Formula:

Given two vectors in terms of unit vectors in component form:

$$\mathbf{A} = 3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

$$\mathbf{B} = -\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$$

Their dot product:

$$\mathbf{A} \cdot \mathbf{B} = (3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}) \cdot (-\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}})$$

Expanding using the unit vector properties:

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} &= (3 \cdot -1)(\hat{\mathbf{i}} \cdot \hat{\mathbf{i}}) + (2 \cdot 4)(\hat{\mathbf{j}} \cdot \hat{\mathbf{j}}) + (-1 \cdot 2)(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}) \\ &= -3(1) + 8(1) - 2(1) \\ &= -3 + 8 - 2 = 3\end{aligned}$$

Thus, the dot product of the two vectors is 3.

Note

The two vectors can alternatively be expressed in column matrix form and their dot product then evaluated as follows:

$$\mathbf{A} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix}$$

Their dot product:

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= \mathbf{A}^T \cdot \mathbf{B} = [3 \quad 2 \quad -1] \cdot \begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix} \\ &= (3 \cdot -1) + (2 \cdot 4) + (-1 \cdot 2) = 3 \end{aligned}$$

Thus, the dot product of the two vectors is 3.

Cross Product of Two Vectors

Introduction

The **cross product**, also known as the **vector product**, is a fundamental operation in vector algebra that arises in three-dimensional space. Unlike the dot product, which results in a scalar quantity, the cross product of two vectors produces a new vector that is **perpendicular** to the plane containing the original vectors. This operation is widely used in physics, engineering, and computer graphics to describe rotational effects, force interactions, and geometric relationships.

Suppose we have two vectors, **A** and **B**, whose **cross product** results in a vector **C** that is perpendicular to both **A** and **B** (see Figure 37). Symbolically, the cross product of vectors **A** and **B**, which produces a perpendicular vector **C**, is written as:

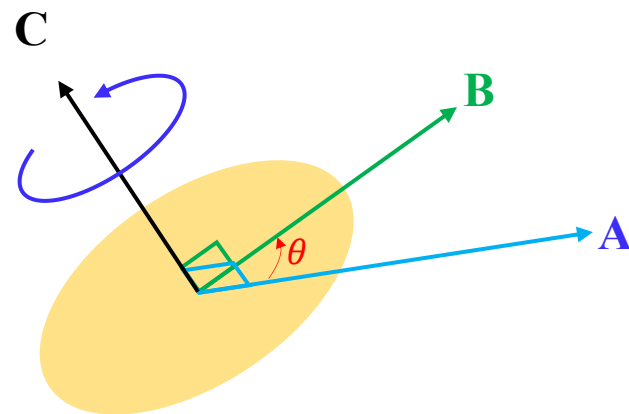
$$\mathbf{C} = \mathbf{A} \times \mathbf{B}$$

where $\times$ is the cross product operator, and the expression is read as:

"C is equal to the cross product of A and B."

The direction of the resulting vector is determined by the **right-hand rule**.

Figure 37



If **A** and **B** are represented as column vectors:

$$\mathbf{A} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix}$$

then the cross product of **A** and **B** that results in the perpendicular vector **C**, is evaluated using the determinant of a 3x3 matrix:

$$\mathbf{C} = \mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix}$$

Note

1. The cross product is only defined for three-dimensional vectors. Attempting to compute the cross product of vectors with dimensions other than three is not valid.

Worked Example

Given two vectors **A** and **B** as:

$$\mathbf{A} = \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 4 \\ 0 \\ -3 \end{bmatrix}.$$

Find $\mathbf{A} \times \mathbf{B}$.

Solution

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -1 & 2 & 5 \\ 4 & 0 & -3 \end{vmatrix}$$

This determinant expands to

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} (2)(-3) - (5)(0) \\ (5)(4) - (-1)(-3) \\ (-1)(0) - (2)(4) \end{vmatrix} = \begin{bmatrix} -6 \\ 17 \\ -8 \end{bmatrix} \blacksquare$$

Applications of the Cross Product in Engineering

The **cross product** of vectors is widely used in engineering fields such as mechanical engineering, electrical engineering, and civil engineering to solve problems related to **torque, force systems, angular momentum, and electromagnetic fields**. It provides a way to calculate a vector that is perpendicular to two given vectors, which is useful for analyzing rotational effects and 3D force interactions.

1. Torque Calculation in Mechanical Engineering

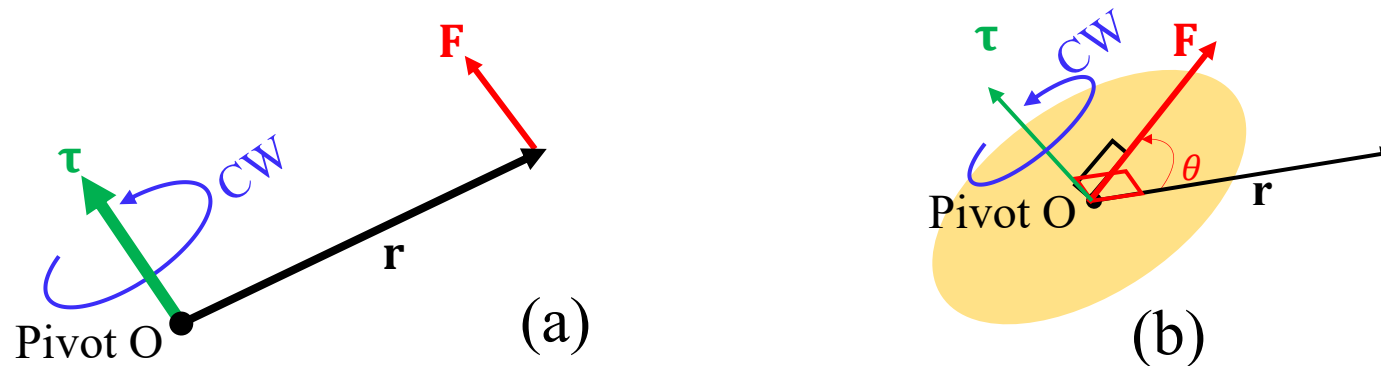
- Torque is the rotational force applied to an object (see Figure 38) and is calculated using the cross product:

$$|\boldsymbol{\tau}| = |\mathbf{r} \times \mathbf{F}| = rF \sin \theta$$

where:

- $\boldsymbol{\tau}$ = Torque vector
- $\mathbf{r}$ = Position vector (from pivot to force application point)
- $\mathbf{F}$ = Force vector

Figure 38



Notes

1. Moment of a Force in Structural Engineering

In structural applications, moments can be classified as:

1. Bending Moment:

- Causes bending in beams or structural elements.
- Example: A horizontal force applied at the end of a cantilever beam.

(see Figure 39)

2. Torsional Moment (Torque):

- A special case where force causes twisting around an axis.
- Example: A shaft transmitting rotational force.

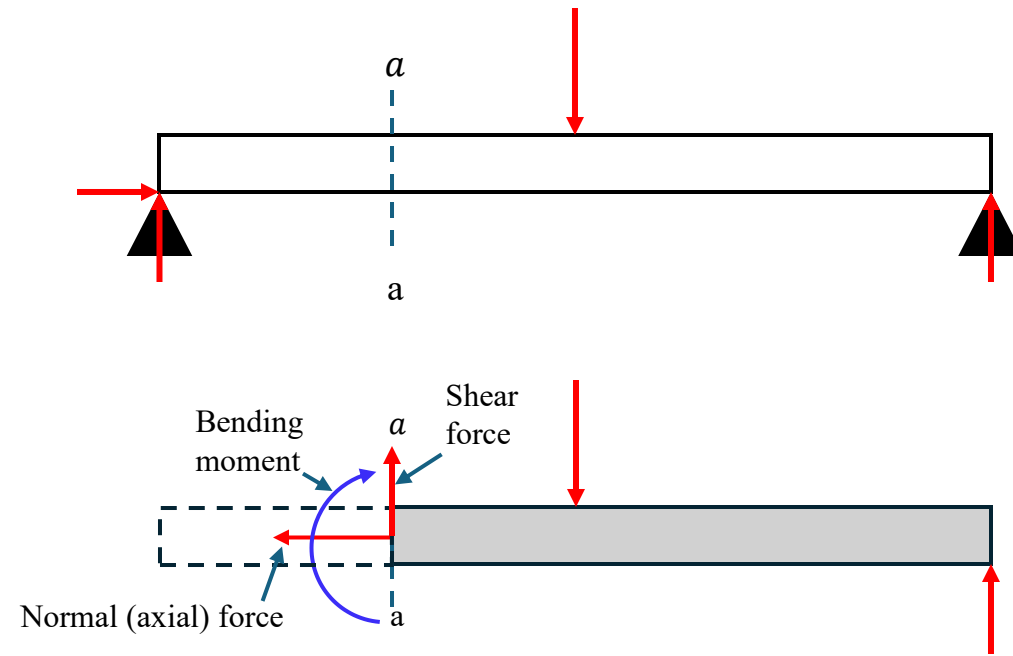


Figure 39

2. Difference between Torque and Moment of a Force

The terms torque and moment of a force are closely related and often used interchangeably, but they have subtle differences based on context and application.

Definition and Conceptual Difference:		
Aspect	Torque	Moment of a Force
Definition	A measure of rotational force causing rotation around an axis or pivot.	A measure of the tendency of a force to cause rotation about an a point or axis.
Context	Primarily used in mechanical systems, especially in rotating objects like engines, gears, etc.	Used in structural analysis and static analysis, such as beams, bridges, and frames.
Application	Rotational motion (e.g., crankshaft of an engine).	Static equilibrium and structural analysis (e.g., beam bending)
Direction	Always about a specific axis	Can be taken about a point, axis, or plane, depending on the problem.
Units	Newtons-meters (N·m)	Newtons-meters (N·m)
Formula	$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$	$\mathbf{M} = \mathbf{r} \times \mathbf{F}$

3. Practical Differences

- **Torque**

- Typically used when forces cause rotation of objects, such as rotating machines, wheels, or engines.
- Example: Applying force to a wrench to rotate a bolt.
- Torque often refers to a rotational effect in **dynamic** systems

- **Moment of a Force**

- Used to describe the tendency of a force to cause bending or rotational effects in structural components.
- Example: A force acting on a cantilever beam causing it to bend.
- Moments are frequently used in **static** equilibrium analysis.

Problem

A 130 N force acts on the end of the 100 cm lever as shown in Figure 40. Determine the moment of the force about O.

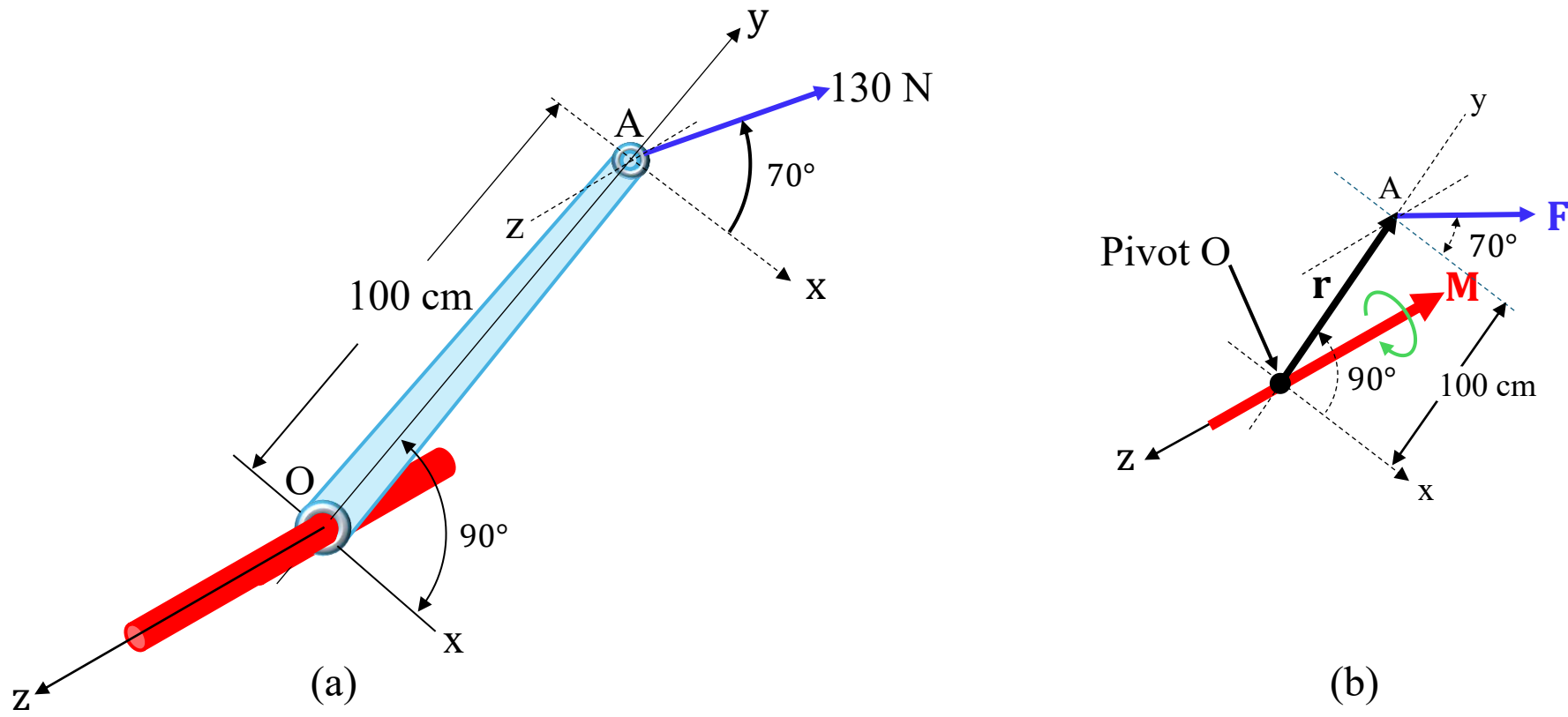


Figure 40

Solution

Step 1: Express position vector $\mathbf{r}$ and force vector $\mathbf{F}$ in column matrix form.

- Expressing the position vector in metres:

$$\mathbf{r} = \begin{bmatrix} 0 \\ 0.9 \\ 0 \end{bmatrix} \text{ m (lever along the y-axis, 0.9 m long)}$$

- Force vector $\mathbf{F}$ is in the x-y plane and inclined at 70° w.r.t the x-axis. In column vector form, we write:

$$\mathbf{F} = \begin{bmatrix} 130\cos 70^\circ \\ 130\sin 70^\circ \\ 0 \end{bmatrix} = \begin{bmatrix} 44.46 \\ 122.16 \\ 0 \end{bmatrix} \text{ N}$$

Step 2: Compute the Moment Using Cross Product

The moment of the force about point O is given by:

$$\mathbf{M} = \mathbf{r} \times \mathbf{F}$$

Using the determinant formula:

$$\mathbf{M} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & 0.9 & 0 \\ 44.46 & 122.16 & 0 \end{vmatrix}$$

Expanding the determinant along the first row using the cofactor method, we get:

$$\mathbf{M} = \hat{\mathbf{i}} \begin{vmatrix} 0.9 & 0 \\ 122.16 & 0 \end{vmatrix} - \hat{\mathbf{j}} \begin{vmatrix} 0 & 0 \\ 44.46 & 0 \end{vmatrix} + \hat{\mathbf{k}} \begin{vmatrix} 0 & 0.9 \\ 44.46 & 122.16 \end{vmatrix}$$

Upon further expansion of the above expression, we finally get:

$$\mathbf{M} = 0\hat{i} + 0\hat{j} - 39.99\hat{k}$$

Thus, the moment about point O is 39.99 in the - z-direction, indicating a clockwise rotation when viewed from the positive-z plane (see Figure 41).

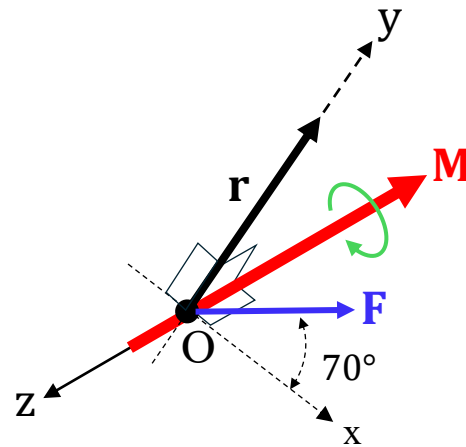


Figure 41

3. Electromagnetic Forces in Electrical Engineering

- The Lorentz force experienced by a charged particle moving in a magnetic field is given by:

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B}) \quad (\text{see Figure 42})$$

where:

- q = Charge
- $\mathbf{v}$ = Velocity vector
- $\mathbf{B}$ = Magnetic field vector

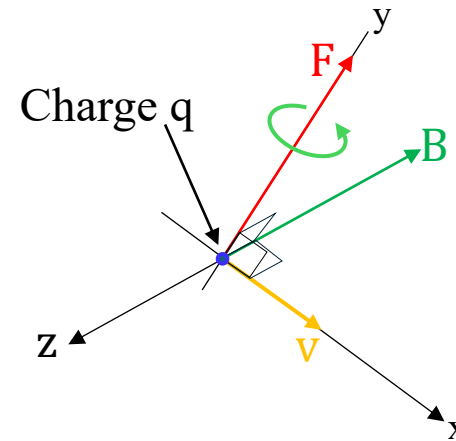


Figure 42

Worked Example

A **charged particle** of charge $q = 2 \times 10^{-6}$ C and mass $m = 5 \times 10^{-6}$ kg moves with a velocity

$$\mathbf{v} = \begin{bmatrix} 4 \\ 2 \\ -3 \end{bmatrix} \text{ m/s}$$

through a constant magnetic field

$$\mathbf{B} = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} \text{ T}$$

Find the acceleration of the particle due to the Lorentz force.

Solution:

The Lorentz force is: $\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$

1. Calculate the cross product

$$\mathbf{v} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 4 & 2 & -3 \\ 1 & -2 & 2 \end{vmatrix} = \begin{bmatrix} -2 \\ -11 \\ -1 \end{bmatrix}$$

2. Calculate the force by the charge:

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B}) = 2 \times 10^{-6} \begin{bmatrix} -2 \\ -11 \\ -1 \end{bmatrix} = \begin{bmatrix} -4 \times 10^{-6} \\ -22 \times 10^{-6} \\ -20 \times 10^{-6} \end{bmatrix} \text{ N}$$

3. Compute the acceleration:

$$\mathbf{a} = \frac{\mathbf{F}}{m} = \frac{1}{5 \times 10^{-6}} \begin{bmatrix} -4 \times 10^{-6} \\ -22 \times 10^{-6} \\ -20 \times 10^{-6} \end{bmatrix} = \begin{bmatrix} -0.8 \\ -4.4 \\ -4 \end{bmatrix} \text{ m/s}^2$$

4. Cross Product as Area Vector of a Parallelogram

In many engineering and physics applications, surfaces are not just geometric objects; they **interact with physical fields**, such as **electric fields, magnetic fields, and fluid flows**. To mathematically describe these interactions, we need a way to represent a surface **quantitatively** and **directionally**—this is where the **area vector** comes in.

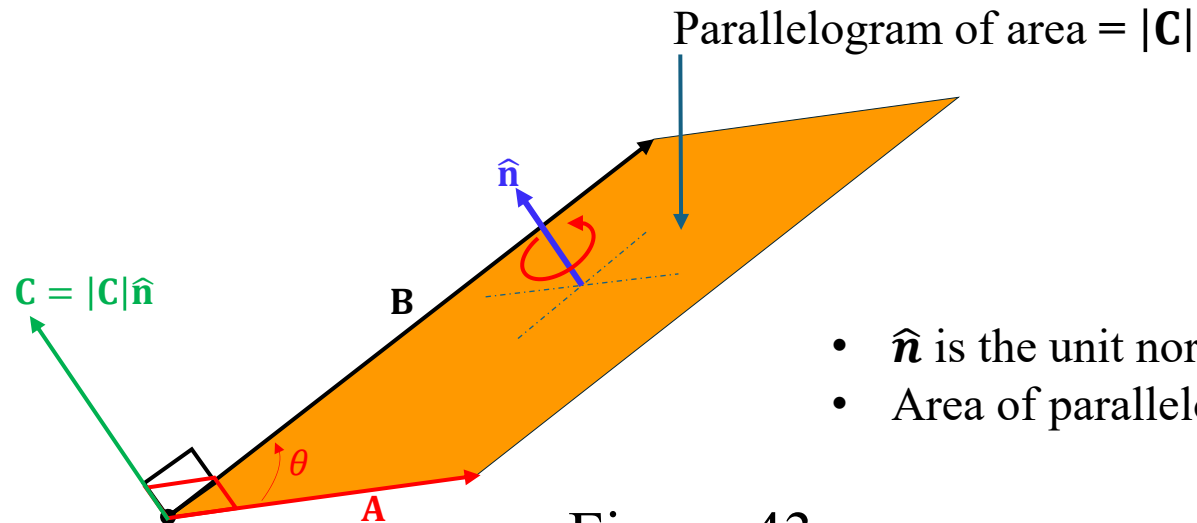
A surface can be represented as a vector with:

- **Magnitude** equal to the area of the surface.
- **Direction** normal (perpendicular to the surface).
- For a **flat surface**, we can determine its **area vector** using the **vector cross product**.

Consider a **parallelogram** defined by edge vectors **A** and **B** (see Figure 43), the **area vector C**, of its surface is given by their cross product:

$$\mathbf{C} = \mathbf{A} \times \mathbf{B} = |\mathbf{C}|\hat{\mathbf{n}}$$

The magnitude of **C** represents the **area** of the parallelogram, while its direction is **perpendicular** to the plane containing **A** and **B**, following the right-hand rule."



- $\hat{\mathbf{n}}$ is the unit normal of the surface of the parallelogram.
- Area of parallelogram, $C = |\mathbf{C}| = |\mathbf{A}||\mathbf{B}|\sin\theta$

Figure 43

Note

The direction of the area vector (resulting cross product) is:

- Perpendicular to the parallelogram, following the right-hand rule, meaning if you curl your fingers from **A** to **B**, your thumb points in the direction of **A** × **B**.
- If the order of vectors is swapped, the direction is reversed:

$$\mathbf{A} \times \mathbf{B} = -(\mathbf{B} \times \mathbf{A})$$

Worked Example

Find the area vector and the unit normal of the slanted surface ABCD of the closed triangular box shown in Figure 44.

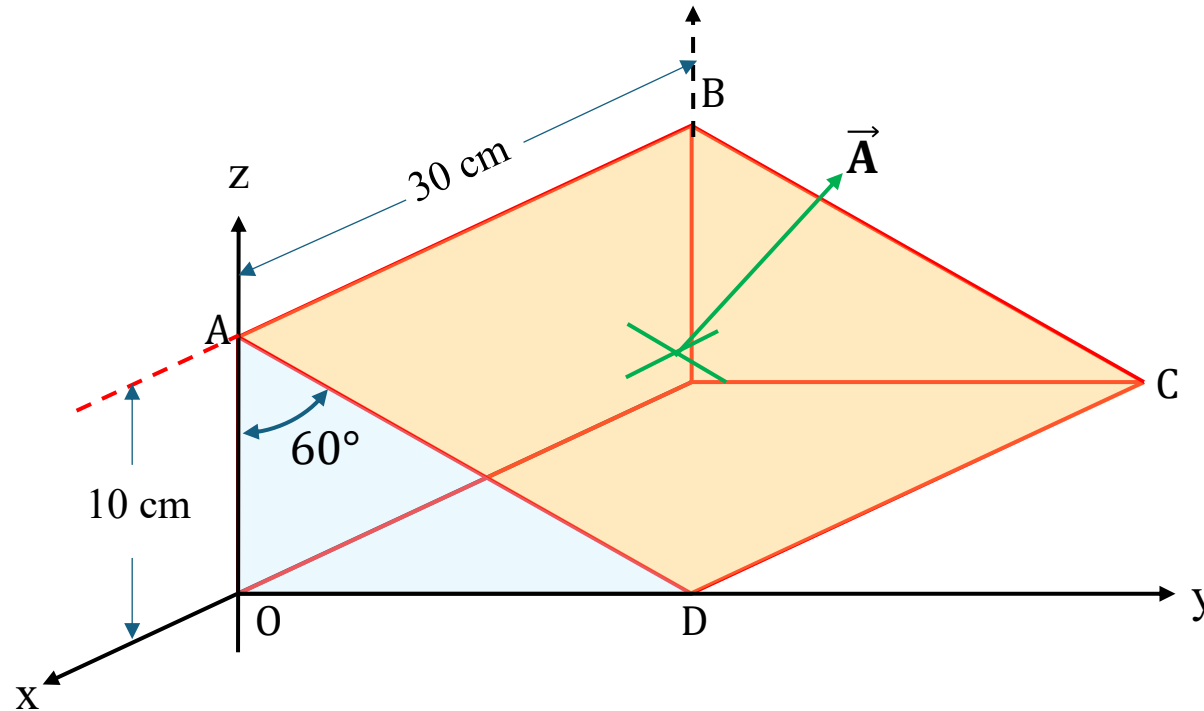


Figure 44

Solution

Let the origin be defined at the bottom left corner of the front triangular side of the prism, as shown in Figure 45.

Length of side OD is :

$$OD = OA \times \tan 60^\circ = 0.1 \times \tan 60^\circ = 0.173 \text{ m}$$

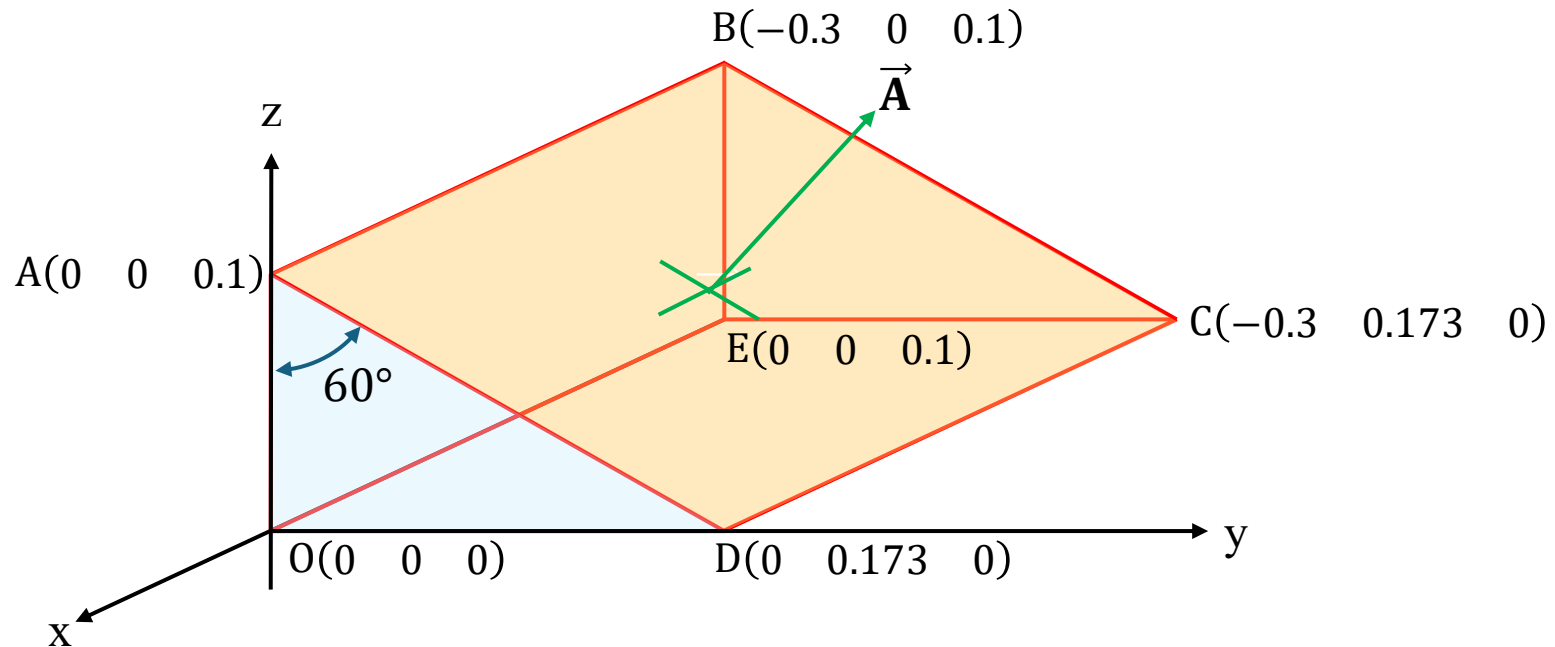


Figure 45

Area vector of the slanted rectangular surface ABCD is obtained by taking the cross product of (displacement) vector $\overrightarrow{AD}$ and vector $\overrightarrow{AB}$; that is

$$\mathbf{A} = \overrightarrow{AD} \times \overrightarrow{AB}$$

Now, from the diagram in Figure 41, we get:

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

where:

$$\overrightarrow{OB} = \begin{bmatrix} -0.3 \\ 0 \\ 0.1 \end{bmatrix} \quad \text{and} \quad \overrightarrow{OA} = \begin{bmatrix} 0 \\ 0 \\ 0.1 \end{bmatrix}$$

Hence,

$$\overrightarrow{AB} = \begin{bmatrix} -0.3 \\ 0 \\ 0.1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0.1 \end{bmatrix} = \begin{bmatrix} -0.3 \\ 0 \\ 0 \end{bmatrix}$$

Also, from the diagram in Figure 41, we get

Now:

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA}$$

where:

$$\overrightarrow{OD} = \begin{bmatrix} 0 \\ 0.173 \\ 0 \end{bmatrix} \quad \text{and} \quad \overrightarrow{OA} = \begin{bmatrix} 0 \\ 0 \\ 0.1 \end{bmatrix}$$

Hence,

$$\overrightarrow{AD} = \begin{bmatrix} 0 \\ 0.173 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.173 \\ -0.1 \end{bmatrix}$$

Hence, we get area vector of the slanted rectangular surface ABCD as:

$$\begin{aligned}\mathbf{A} = \overrightarrow{AD} \times \overrightarrow{AB} &= \begin{bmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & 0.173 & -0.1 \\ -0.3 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} -0.017 \\ 0.03 \\ -0.1 \end{bmatrix} = 0.106 \begin{bmatrix} -0.161 \\ 0.284 \\ -0.945 \end{bmatrix} m^2\end{aligned}$$

The magnitude of the area vector $\mathbf{A}$ is determined by computing its modulus:

$$\text{Area} = |\mathbf{A}| = 0.106 \text{ m}^2$$

The unit normal vector $\hat{\mathbf{n}}$ to the area is determined by normalizing the area vector.

This is done by dividing the area vector $\mathbf{A}$ by its magnitude $|\mathbf{A}|$:

$$\hat{\mathbf{n}} = \frac{\mathbf{A}}{|\mathbf{A}|} \quad (\text{see Figure 46})$$

Thus, we get:

$$\hat{\mathbf{n}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{1}{0.106} \begin{bmatrix} -0.017 \\ 0.03 \\ -0.1 \end{bmatrix} = \begin{bmatrix} -0.161 \\ 0.284 \\ -0.945 \end{bmatrix}$$

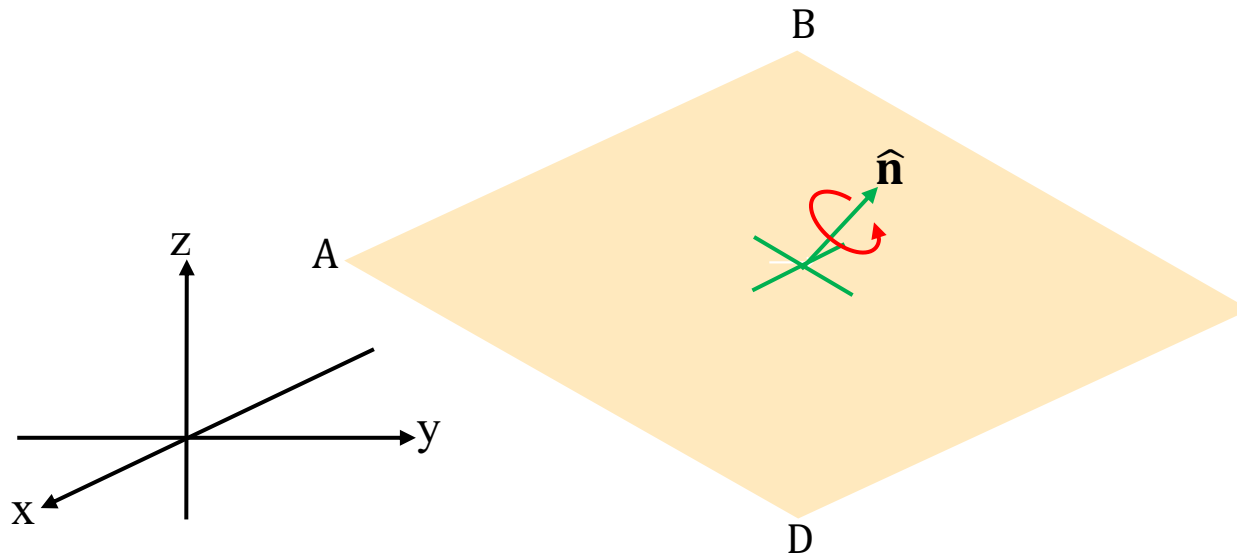


Figure 46

Properties of the Cross Product

The **cross product** of two vectors in three-dimensional space has several important properties that make it useful in engineering, physics, and mathematics.

1. Anti-commutative Property (Skew-Symmetry)

$$\mathbf{A} \times \mathbf{B} = -(\mathbf{B} \times \mathbf{A})$$

- Swapping the order of the vectors reverses the direction of the resulting vector.
- Example: If $\mathbf{A} = (1, 0, 0)$ and $\mathbf{B} = (0, 1, 0)$, then:

$$\mathbf{A} \times \mathbf{B} = (0, 0, 1)$$

$$\mathbf{B} \times \mathbf{A} = (0, 0, -1)$$

2. Distributive Property (Over Vector Addition)

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$$

- The cross product distributes over vector addition.

- **Example:**

- If $\mathbf{A} = (1, 0, 0)$, $\mathbf{B} = (0, 1, 0)$, $\mathbf{C} = (0, 0, 1)$, then

$$\begin{aligned}\mathbf{A} \times (\mathbf{B} + \mathbf{C}) &= (0, 0, 1) + (0, -1, 0) \\ &= (0, -1, 1)\end{aligned}$$

3. Scalar Multiplication Property

$$(c\mathbf{A}) \times \mathbf{B} = c(\mathbf{A} \times \mathbf{B})$$

$$\mathbf{A} \times (c\mathbf{B}) = c(\mathbf{A} \times \mathbf{B})$$

- Scaling either vector scales the result by the same factor.
- **Example:**
 - If $\mathbf{A} = (2, 0, 0)$, $\mathbf{B} = (0, 3, 0)$, $c = 5$:

$$(5\mathbf{A}) \times \mathbf{B} = 5(6\hat{\mathbf{k}}) = (30\hat{\mathbf{k}})$$

4. Orthogonal Property

$$\mathbf{A} \times \mathbf{B} \perp \mathbf{A}, \quad \mathbf{A} \times \mathbf{B} \perp \mathbf{B}$$

- The cross product is perpendicular to both original vectors.

5. Magnitude Property

$$|\mathbf{A} \times \mathbf{B}| = |\mathbf{A}||\mathbf{B}|\sin\theta$$

- The magnitude of the cross product depends on the sine of the angle between the two vectors.
- If $\theta = 0^\circ$ or 180° , then $\sin \theta = 0$, and the cross product is zero (parallel vectors).

6. Zero Vector Property

$$\mathbf{A} \times \mathbf{A} = \mathbf{0}$$

- The cross product of any vector with itself is the zero vector, because the sine of the angle (0 degrees) is zero.
- **Example:**
 - If $\mathbf{A} = (2, 3, 4)$, then:

$$\mathbf{A} \times \mathbf{A} = \mathbf{0}$$

7. Cross Product of Standard Unit Vectors

The standard unit vectors $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ follow cyclic relationships:

$$\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}, \quad \hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}, \quad \hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$$

$$\hat{\mathbf{j}} \times \hat{\mathbf{i}} = -\hat{\mathbf{k}}, \quad \hat{\mathbf{k}} \times \hat{\mathbf{j}} = -\hat{\mathbf{i}}, \quad \hat{\mathbf{i}} \times \hat{\mathbf{k}} = -\hat{\mathbf{j}}$$

This cyclic property is useful in computations involving coordinate systems.

8. Triple Products Involving the Cross Product

1. Vector Triple Product:

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$$

- Used in mechanics and physics to simplify force and moment calculations.

2. Scalar Triple Product:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$$

- Results in a scalar and represents the volume of a parallelepiped formed by three vectors.

Summary of Properties:

1. **Antic-commutative:** $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$
2. **Distributive:** $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$
3. **Scalar multiplication:** $(c\mathbf{A}) \times \mathbf{B} = c(\mathbf{A} \times \mathbf{B})$
4. **Orthogonality:** $\mathbf{A} \times \mathbf{B}$ is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.
5. **Magnitude formula:** $|\mathbf{A} \times \mathbf{B}| = |\mathbf{A}||\mathbf{B}|\sin\theta$
6. **Zero vector:** $\mathbf{A} \times \mathbf{B} = \mathbf{0}$
7. **Unit vector identities:** Cyclic relationships between $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$
8. **Triple products:** Useful for volume and dimensional analysis.

Scalar Triple Product

The **scalar triple product** of three vectors **A**, **B**, and **C** is a mathematical operation that provides a scalar quantity representing the **signed volume** of the parallelepiped formed by the three vectors.

1. Definition of the Scalar Triple Product

The scalar triple product is defined as:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$$

where:

- $\mathbf{B} \times \mathbf{C}$ results in a vector that is perpendicular to the plane containing **B** and **C**.
- The dot product with **A** projects **A** onto this perpendicular direction, yielding a scalar.

2. Geometric Interpretation

The magnitude of the scalar triple product:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$$

represents the volume of the parallelepiped formed by the three vectors.

- If the result is positive, the vectors form a right-handed system.
- If the result is negative, the vectors form a left-handed system (opposite orientation).
- If the result is zero, the vectors are coplanar, meaning they lie in the same plane, and the volume is zero.

3. Determinant Formulation

The scalar triple product can be computed using the determinant of a matrix form by the components of the vectors:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

If the determinant is zero, the vectors are linearly dependent (coplanar).

4. Properties of the Scalar Triple Product

1. Cyclic Permutation:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

2. Antisymmetric Property (Swapping any two vectors negates the result):

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = -\mathbf{A} \cdot (\mathbf{C} \times \mathbf{B})$$

3. Zero Condition:

If the scalar triple product is zero, the vectors are coplanar, meaning they lie in the same plane.

4. Distributive Property:

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) \times \mathbf{D} = \mathbf{A} \cdot \mathbf{B}(\mathbf{C} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D})$$

5. Applications in Engineering and Physics

1. Volume Calculation:

- Used to compute the volume of parallelepiped in physics and engineering.

2. Coplanar Check

- To determine if three vectors lie in the same plane.

3. Rigid Body Mechanics:

- Applied in moment calculations and force systems.

4. Electromagnetism:

- Appears in expressions involving torque, magnetic flux, and force analysis.

Example Application: Finding the Volume of a Parallelepiped

Let **A**, **B** and **C** be vectors in R^3 that are not planar and that form a three-dimensional parallelepiped as shown in Figure 47. It can be shown that the volume of the parallelepiped is given by the modulus of the scalar triple product. That is:

$$\text{volume} = |(\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}| = Ah$$

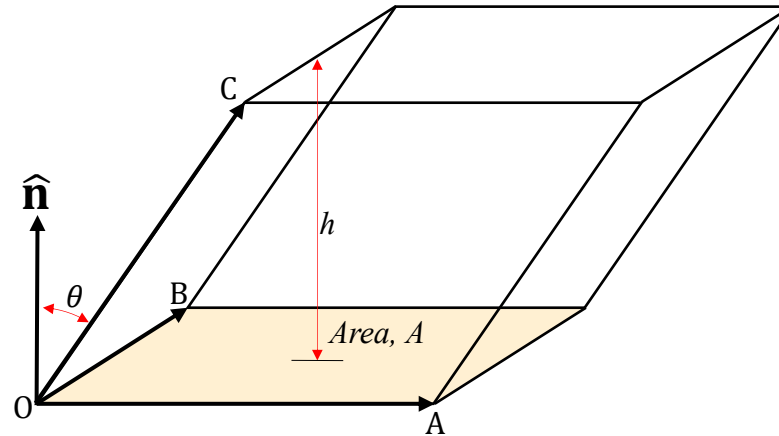


Figure 47

Worked Example

Find the volume of the parallelepiped formed by the adjacent edge vectors

$$\mathbf{A} = \langle 1 \quad 3 \quad 5 \rangle,$$

$$\mathbf{B} = \langle 2 \quad -1 \quad 0 \rangle,$$

$$\mathbf{C} = \langle -3 \quad 0 \quad -1 \rangle.$$

Solution

The parallelepiped formed by the three adjacent vectors is shown (not to scale) in Figure 48.

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 3 & 5 \\ 2 & -1 & 0 \end{vmatrix} = \begin{bmatrix} 3 \\ 10 \\ -7 \end{bmatrix}$$

Hence,

$$\begin{aligned} (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C} &= [3 \quad 10 \quad -7] \begin{bmatrix} 3 \\ 10 \\ -7 \end{bmatrix} \\ &= -1 \end{aligned}$$

Therefore, volume of parallelepiped is:

$$\text{volume} = |(\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}| = |-1| = 1 \text{ unit}^2 \blacksquare$$

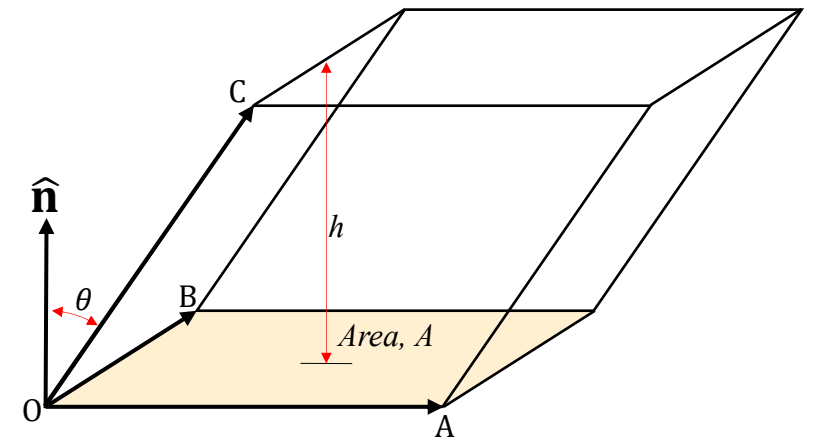


Figure 48

END OF UNIT 2